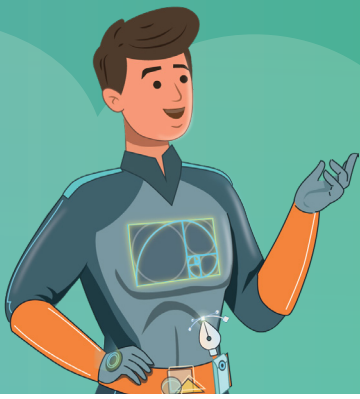




Character Animation

10

Hello DeZain! I'm wondering whether I can animate this 3D character.

Absolutely! It can be done through character animation. It is the process of bringing a character to life through movement and expression. It involves creating a sequence of poses and movements that convey emotions, actions, and interactions, giving the character a sense of personality and realism.

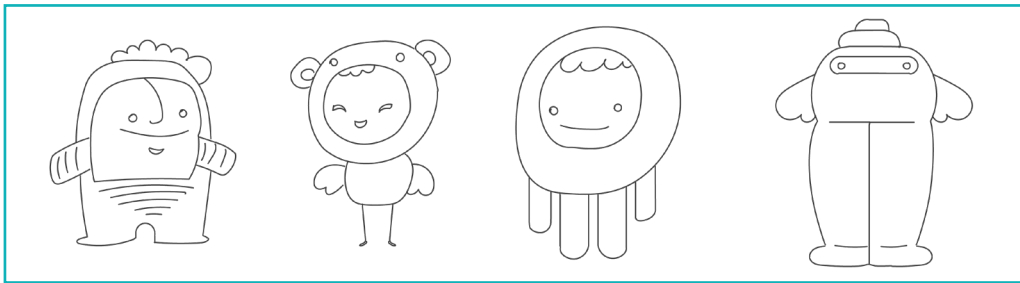


Character animation refers to the process of bringing a character to life through movement. It involves creating and manipulating a character's actions, expressions, and behaviours to make them appear lifelike and to give them their personality. Character animation is widely used in various forms of media, such as films, video games, advertisements, and animated series.

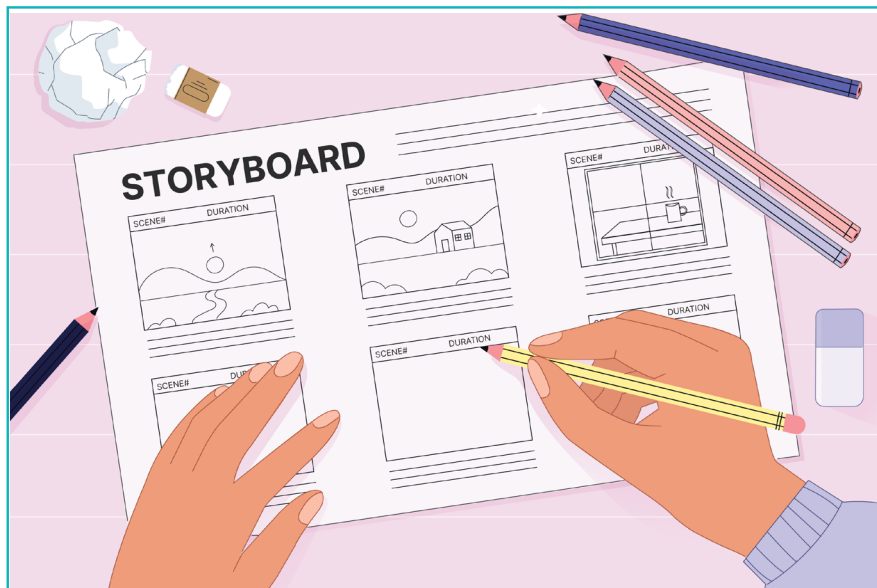
Traditionally, character animation relied on the manual process of meticulously drawing each frame, which requires great attention to detail and a high level of skill. However, as technology advanced, computer animation has gained popularity, and nowadays, character animation is achieved using computer software and various techniques such as 3D modelling, rigging, and keyframing.

The process of character animation typically involves several key steps:

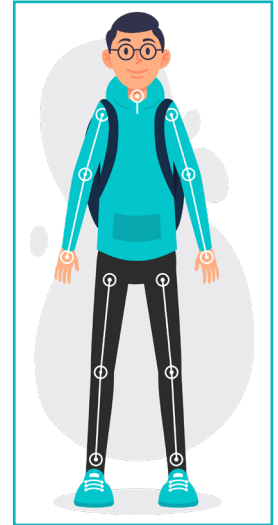
- a. Character Design:** The first step is to create a character. It includes designing its appearance, identifying its physical attributes, and establishing its personality and traits. The design should be visually appealing and capable of conveying emotions and movements effectively.



- b. Storyboarding:** Storyboarding is the process of planning out the sequence of actions and movements that the character will perform. It involves creating a series of rough sketches or storyboard panels to outline the key poses and actions required for the animation.



c. Rigging: Rigging is the process of creating a digital skeleton or structure for the character. It involves defining a hierarchical system of joints and bones that can be used to control the character's movement. The rigging process also includes setting up controls for the character's facial expressions and other deformations.

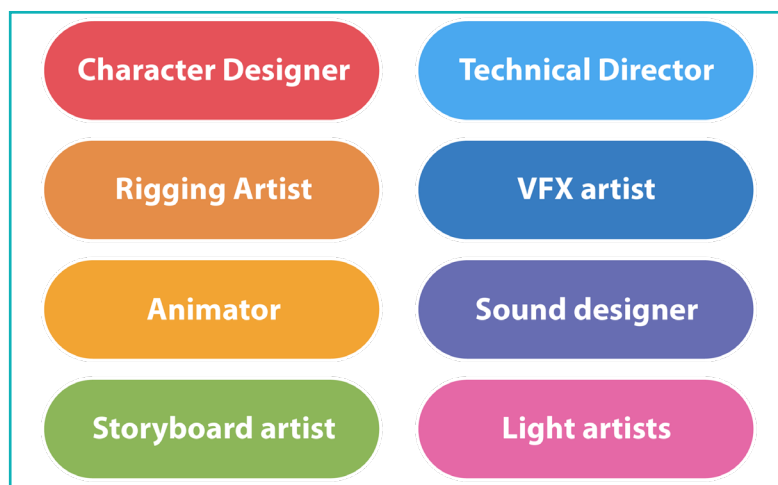


d. Keyframing: It is a technique used to define the important poses and movements in animation. The animator creates a series of keyframes, which are the significant positions or moments in the animation, and then fills in the in-between frames to create a smooth motion. Keyframes help establish the timing, spacing, and overall movement of the character.

In addition to the above-mentioned steps, the complexity of a project may introduce additional stages, such as:

- Adjusting the timing and spacing in the character's actions.
- Smooth transitions and believable movements between keyframes.
- Fine-tuning the details to enhance the overall quality of the animation.
- Rendering it into a final output format, depending on the scale of the project.

The process of character animation involves a team of skilled professionals. They perform the above-mentioned tasks, offer their skills, and collaborate to bring the characters to life. Here are some of the key people involved in the process:



Activity

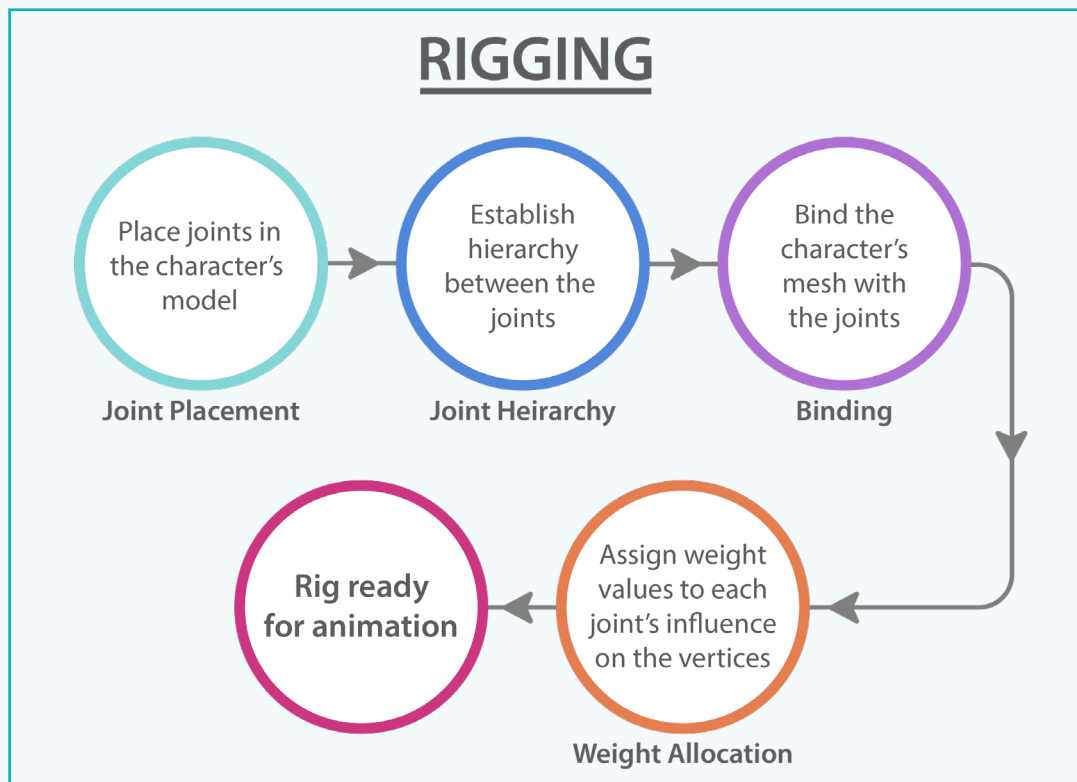
In the previous lesson, we created the model of a character, which is the first step in the process of Character Animation.

The next step is Storyboarding. For learning purposes, we do not need to create an elaborate storyboard. We will simply create a short animation clip showing the character hopping up and down in a loop.

After storyboarding is the process of Rigging. Before starting it, let us understand it in detail.

Rigging

Rigging is a critical step in character animation because a well-designed rig enables smooth and realistic movement of the character. This, in turn, involves the following steps:



Depending on the complexity of the character's design, there may be more steps involved, like Control Setup, IK/FK Switching, Facial Rigging, and Deformation Refinement.

A well-designed and properly executed rig is essential for animators to bring characters to life with natural and expressive movements.

Activity

Create a rig for the model.

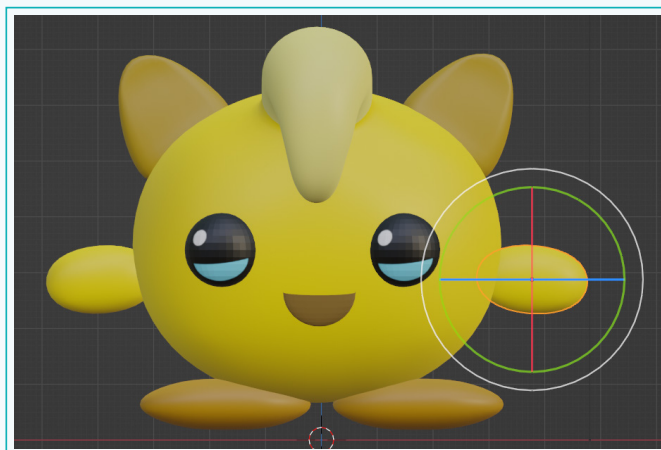
Let us begin the process of rigging with the first step in the rigging process, which is Joint Placement for the character model that we created in the previous lesson.

Below is the model created in the previous lesson. It consists of different shapes that make up the hands, feet, ears, and body of the model.



- 1 Rotate the arms horizontally to place the character in the T-Pose.

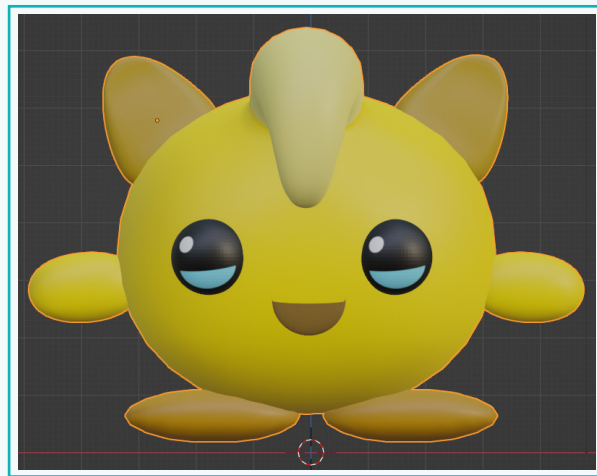
The **T-pose** allows the character model to be in a symmetrical position, with arms outstretched and legs together. This symmetry simplifies the process of creating a symmetrical rig.



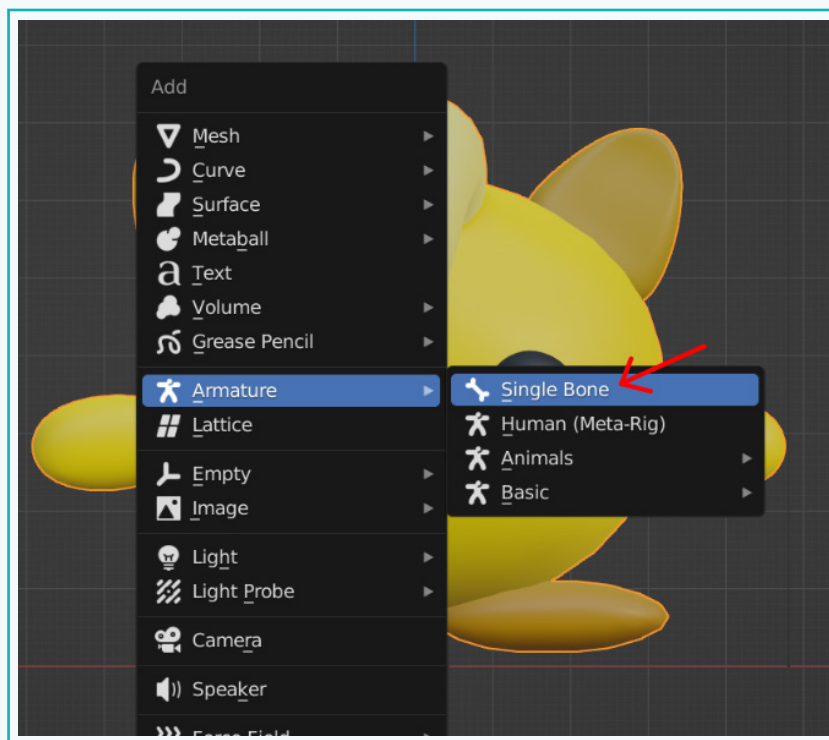
Activity

Let us join them all to turn them into one form of a single mesh.

- 2 Press 'A' to select all and then press 'Ctrl + J' to join them together.

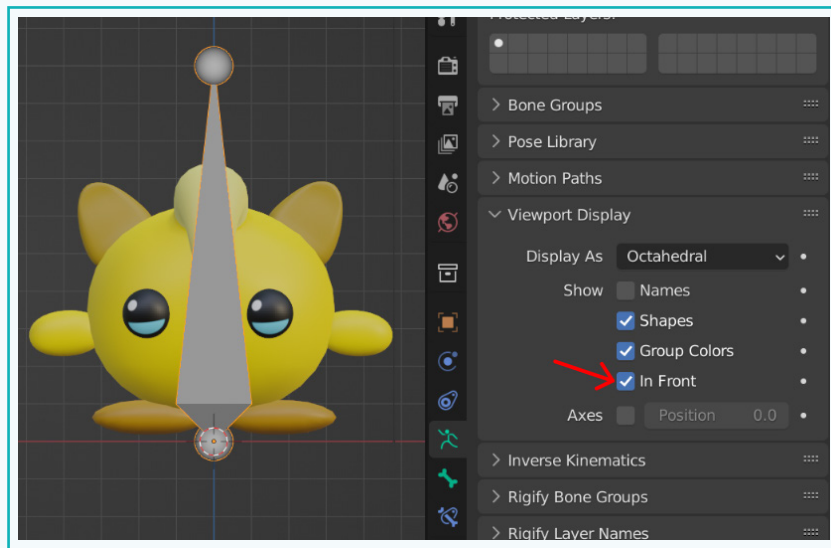


- 3 For creating joints, stay in the object mode and press 'Shift + A' → 'Armature' → 'Single Bone'.

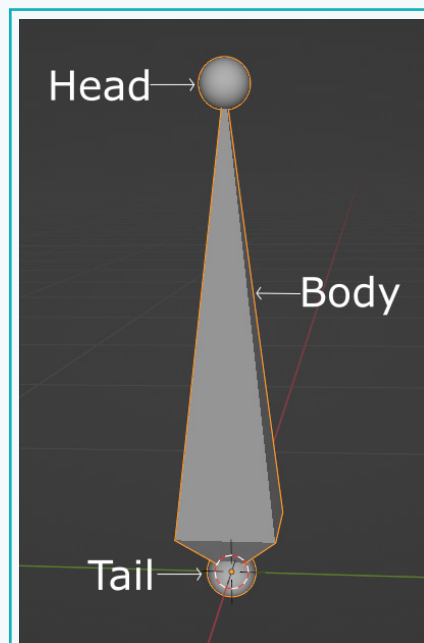


Activity

- 4 If the bone is not visible, go into the 'Object Data Properties' window and under 'Viewport Display', enable the 'In Front' option. The bone becomes visible in front of the object.

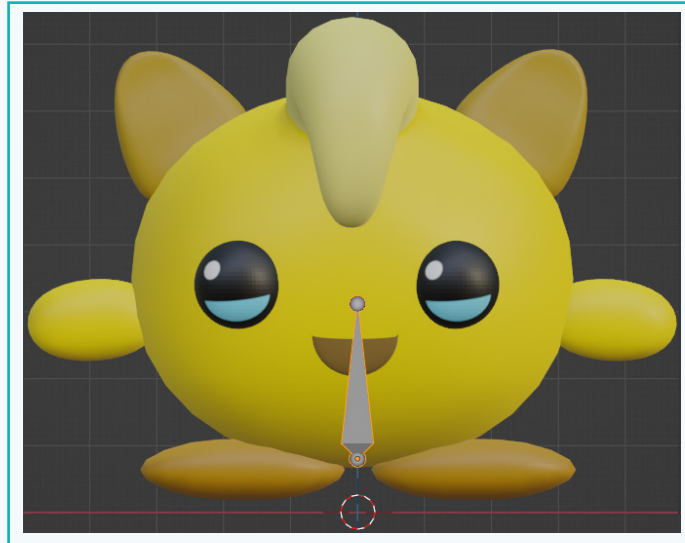


In Blender, the bones are made up of three parts – the head, body, and tail. The head and tail can be moved, scaled, and extruded, as required. The body is what connects the two.

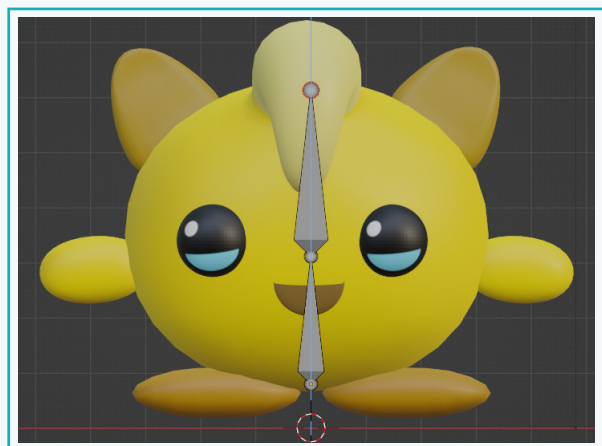
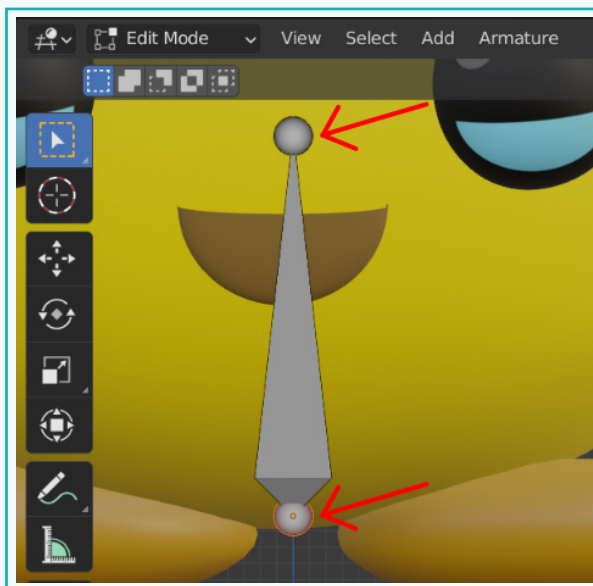


Activity

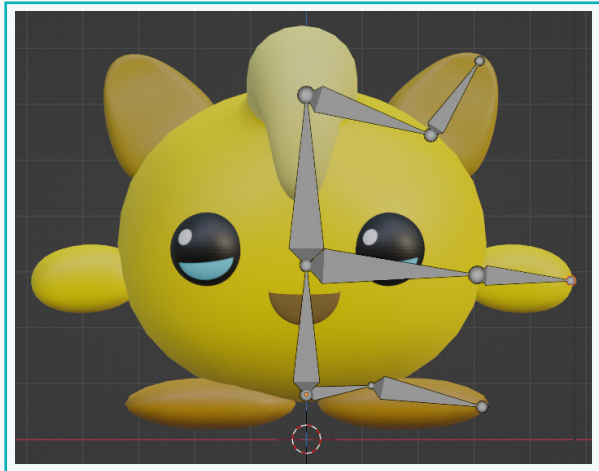
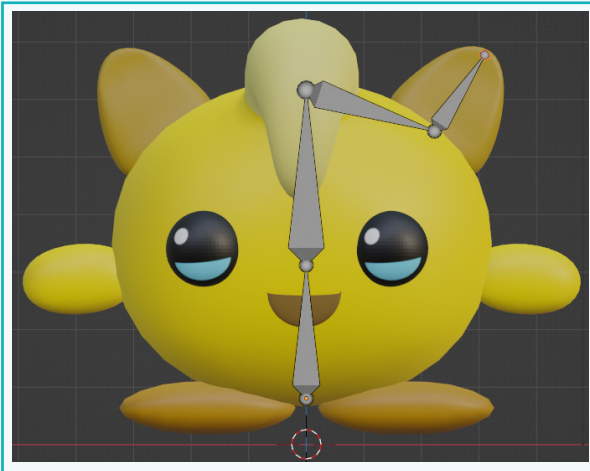
- 5 Press 'S' to scale down the bone to approximately half the height of the character.



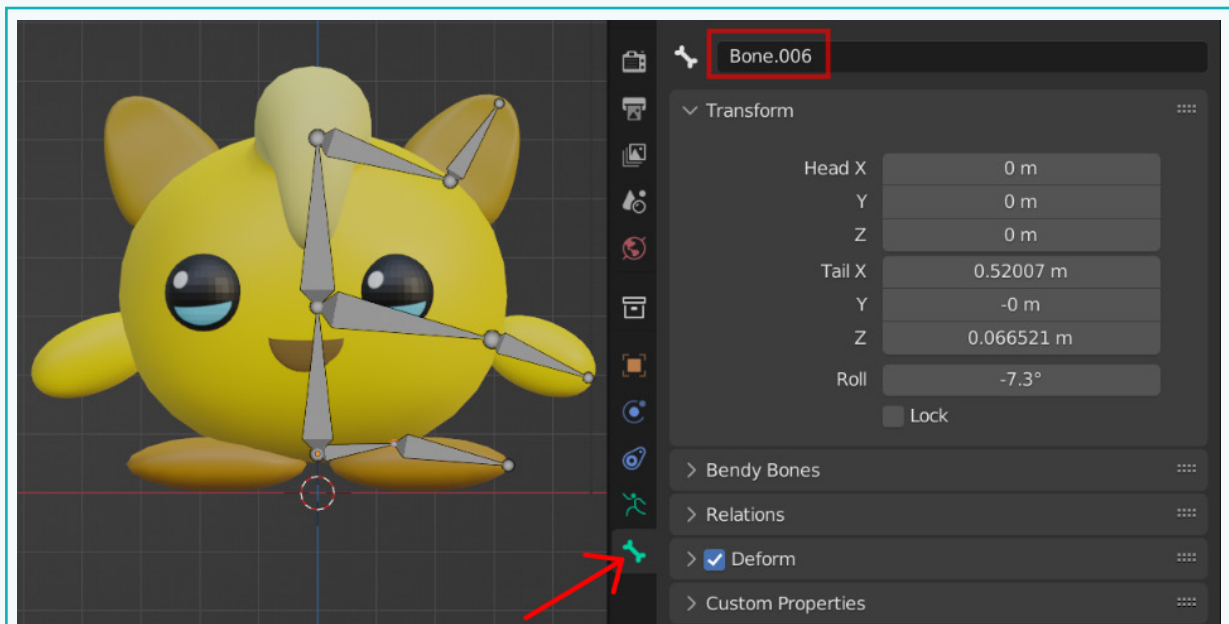
- 6 Switch to edit mode and select the head or the tail of the bone. Press E and extrude the bone sideways/upwards to create a skeletal structure of one side of the character's body as shown.



Activity



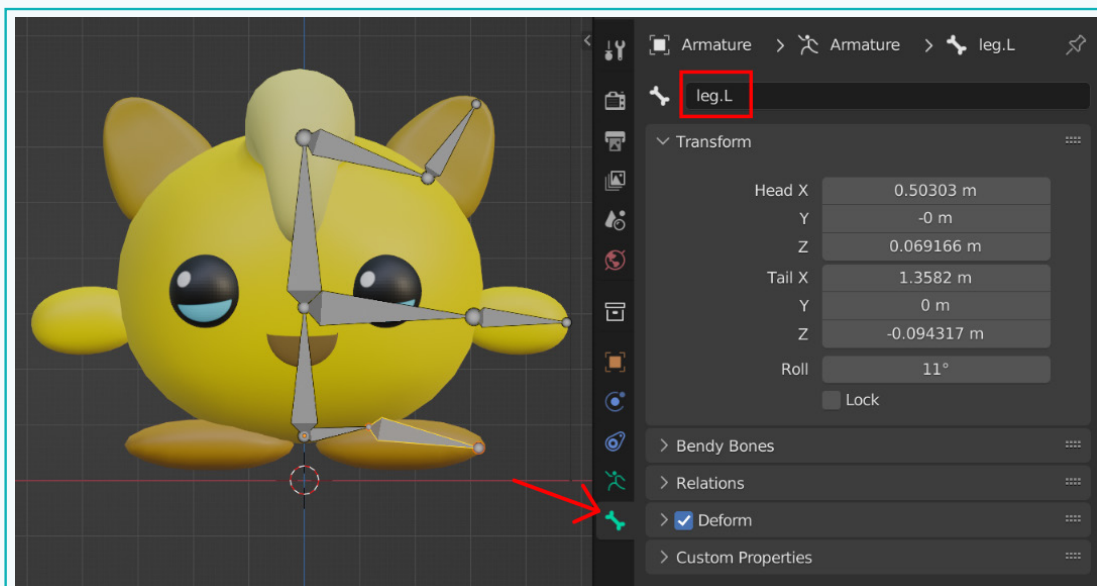
Go into the 'Edit Mode' and select any bone. Click on the 'Bone' properties. You will see that the bone has been allotted a name by Blender. This name is not very intuitive, and it is difficult to figure out which part of the body the bone belongs to.



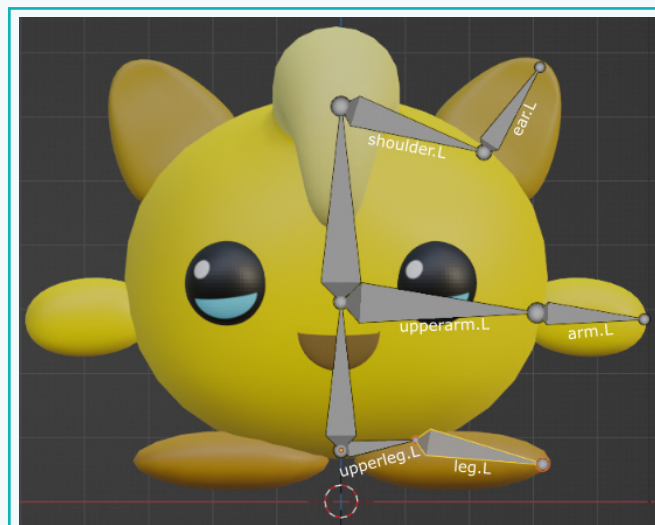
Activity

- 7 Let us rename the bones to relate which bone belongs to which part of the body.

Make sure that the naming convention is in the syntax – ‘nameofbone.L’, where ‘L’ stands for ‘Left’. Doing this will allow us to use the ‘Symmetrize’ tool in the next step to create the skeletal structure on the other side. Also, Blender will automatically name the bones on the other side using the same naming convention.

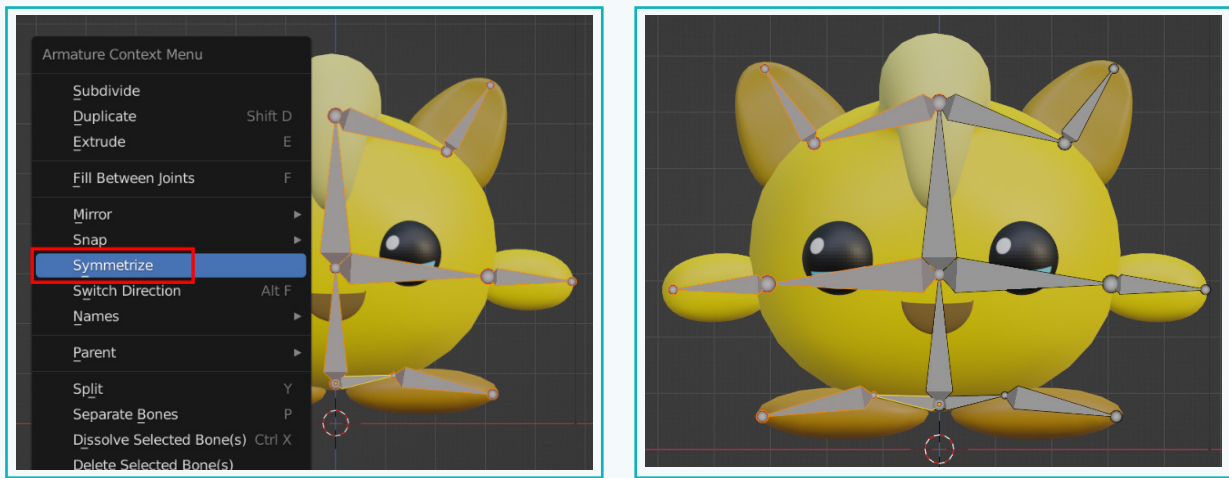


- 8 Select each bone and rename them. The below naming convention can be followed.



Activity

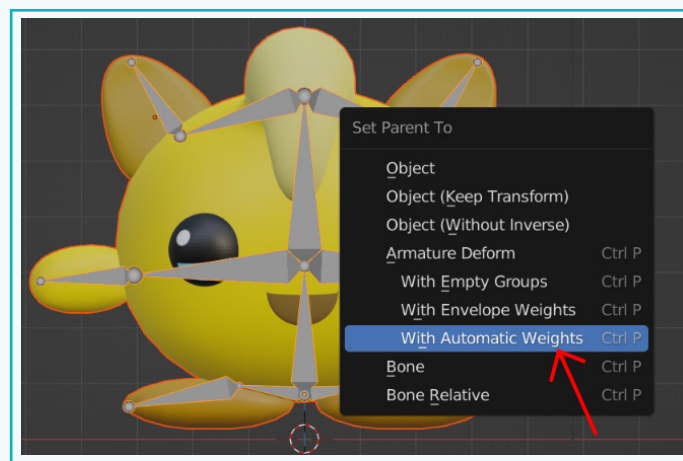
- 9 Stay in the 'Edit' mode. Press 'A' to select all, right-click → 'Symmetrize'. The bone structure for the other side gets created.



The armature for our model has been set up. As we have a very uncomplicated bone structure and our animation will also be simple, we do not need to perform a Joint hierarchy setting.

Let us continue with the **Binding** step, which involves associating the character's mesh with the joints. The character's vertices are assigned to specific joints of the armature, which are represented by vertex groups. This determines which joints influence the movement of which vertex.

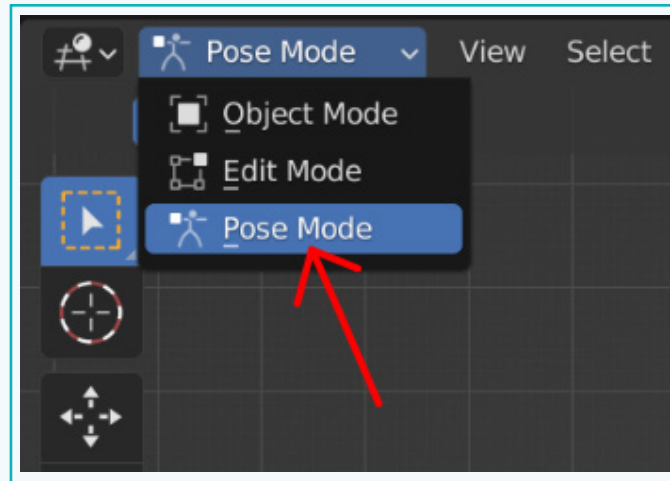
- 10 Select the mesh, then shift-click the armature. Press 'Ctrl + P' for the 'Set Parent To' menu and select 'With Automatic Weights'.



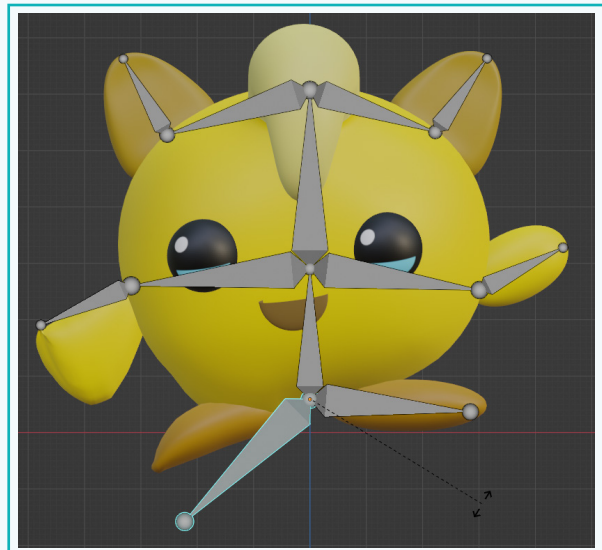
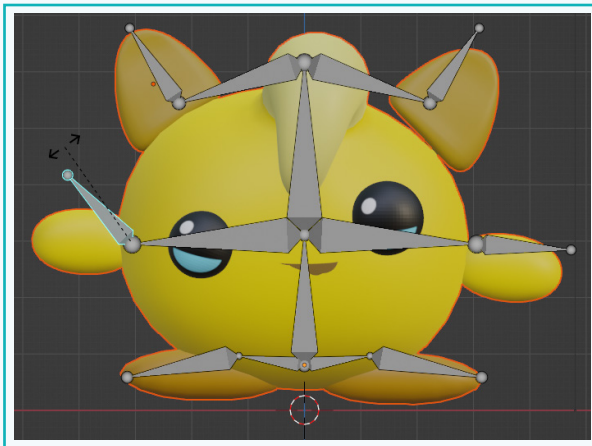
This connects the armature to the mesh.

Activity

- 11 Select the armature and go into the 'Pose' mode.



- 12 Select any bone and move it. Parts of the mesh will get deformed when the bone moves.



Activity

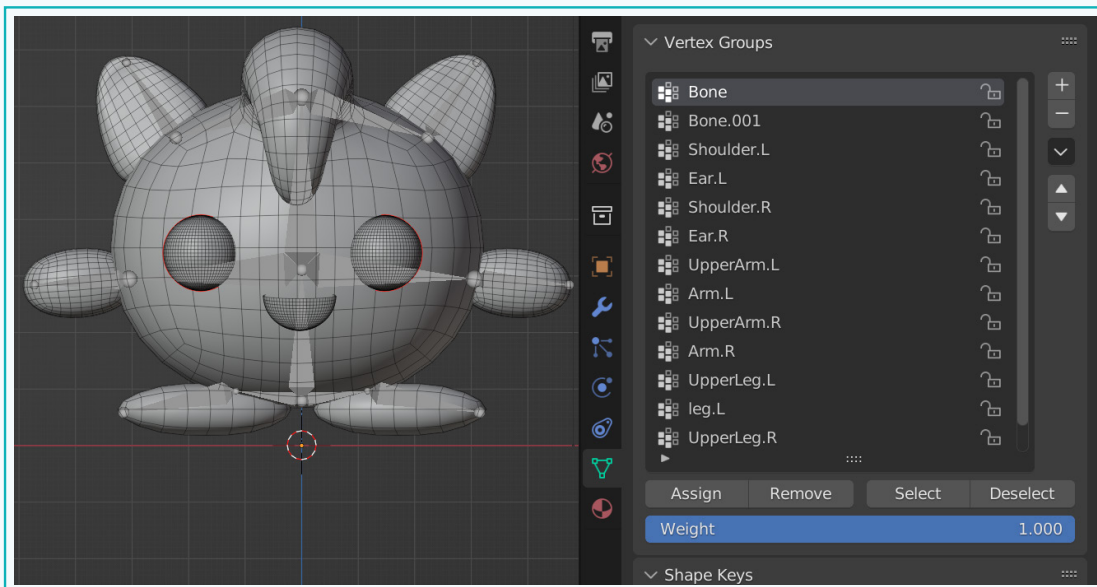
- 13 To clear all the transforms and reset the mesh and the bones to their original shape, press 'F3' and type "Reset" in the search box. Select the 'Pose' → 'Clear Transform'.

During the process of 'Binding', Blender tries to allocate specific vertices to specific bones. This might not always be correct, and we will be required to manually allocate the weights to ensure correct rigging and animation.

The process of assigning weight values to each joint's influence on the vertices is known as **Weight Painting or Weight Allocation**. The weight values of the vertices determine how much a joint/bone affects them.

For example, if a shoulder joint should move the arm but not the head, the shoulder joint would have a high weight value for the arm vertices and a low weight value for the head vertices.

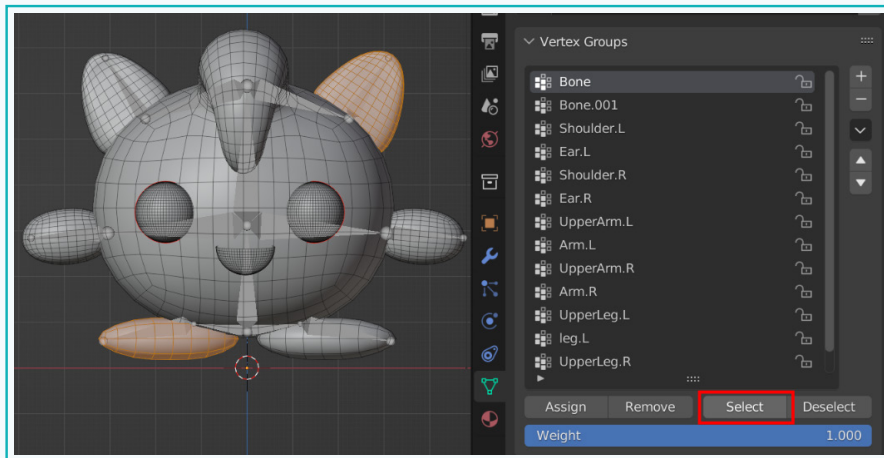
- 14 We can see which vertices are affected by which bone by selecting the mesh and going into the 'Edit' mode. Go into the 'Object Data Properties' window and the list of vertex groups is displayed here.



- 15 To see which vertices are affected by which bone, click on the name of the bone/vertex group and then click on the 'Select' button at the bottom of the list.

Activity

The vertices allocated to the selected vertex group/joint get selected. As can be seen, the vertex allocation has not taken place accurately. For the spine bone, the vertices on the left ear and right leg are getting affected.

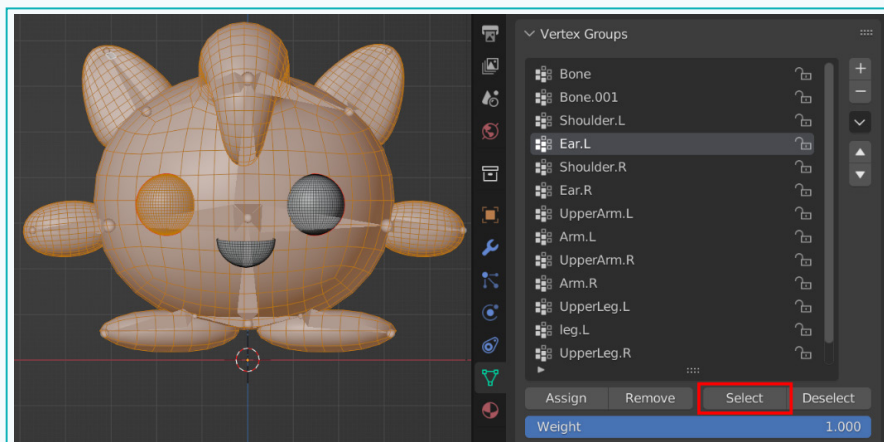


- 16 Go into the 'Object' mode and select the armature. Go into the 'Pose' mode and try to move the spine bone. We will notice that the left ear and right leg end up moving due to this incorrect vertex allocation.

We can check the vertex allocation for the other bones as well in this way.

Note:

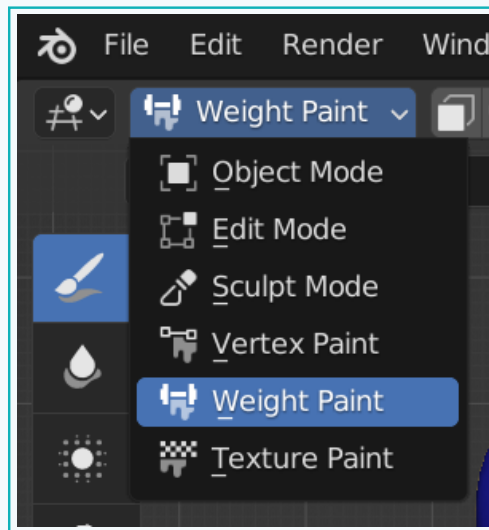
Before we select the vertices of the next bone, we need to make sure to deselect the previous bone vertices by clicking anywhere in the 3D workspace. Otherwise, the previous vertices get added to the next selection.



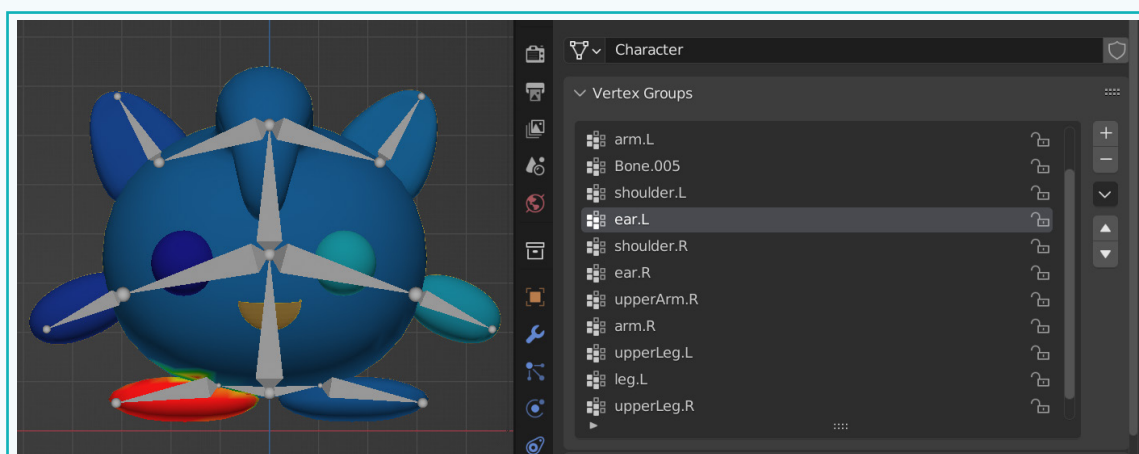
Activity

We can also check the allocation of weights to the vertices in the 'Weight Paint' mode.

- 17 Select the mesh and select the 'Weight Paint' mode. In the 'Weight Paint' mode, the weights are represented by colours.



Each colour represents the amount of influence or weight the vertex group has over specific vertices. E.g., blue represents a weight of '0' or no effect, whereas red represent a weight of '1' or '100%' or the maximum influence that a vertex group can have. Green represents '0.5' or '50%' weight, while the other colours are simply a blend of colours between these three, with appropriate weight distribution.



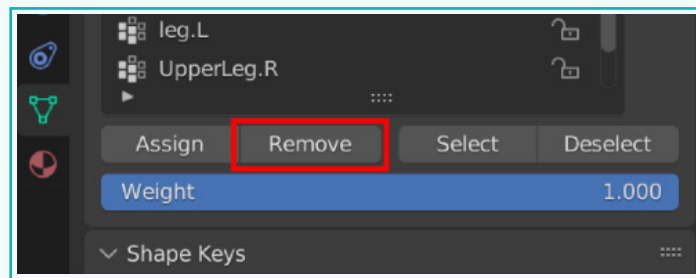
The incorrect allocation of vertices can be addressed using either the 'Weight Painting' method or by precisely allocating weightage to the vertices.

Activity

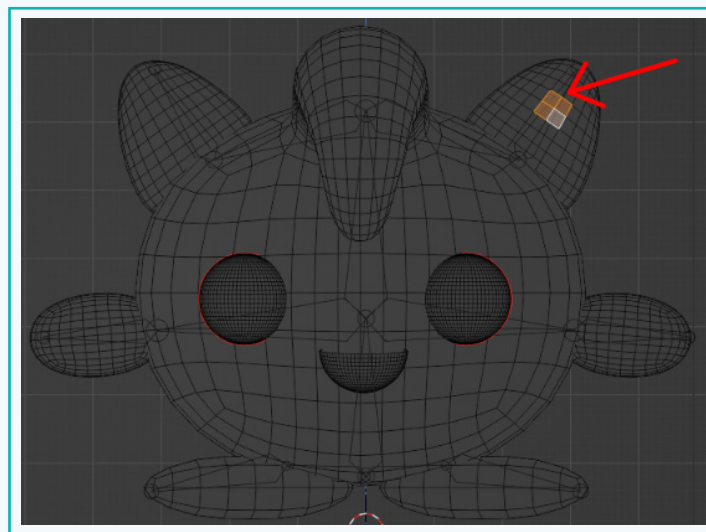
Weight painting can be a challenging process for beginners since we need to have a good understanding of the underlying anatomy of the character, have good brush control, and pay attention to detail. It is often an iterative process in which we must test the weights in different poses, adjust, and repeat until we have achieved the desired results.

To understand weight allocation in a better way, let us precisely allocate weightage to the vertices.

- 18 Select the left ear vertex group (Ear.L) from the list. Click on 'Remove'. This will remove all the vertices present in the left ear vertex group. We can verify this by clicking on the 'Select' button after deselecting the vertices first. No vertex should get selected. Now, we can decide which vertices should go into the vertex group.

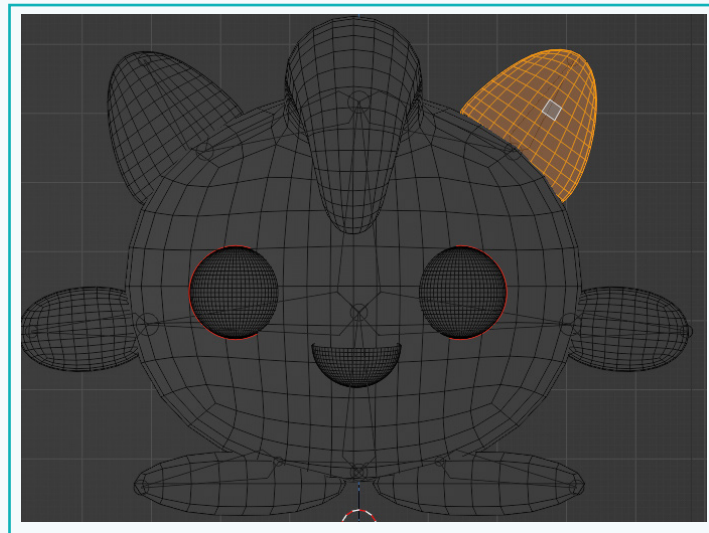


- Now, select one or more vertex/edges/faces on the left ear, as shown.



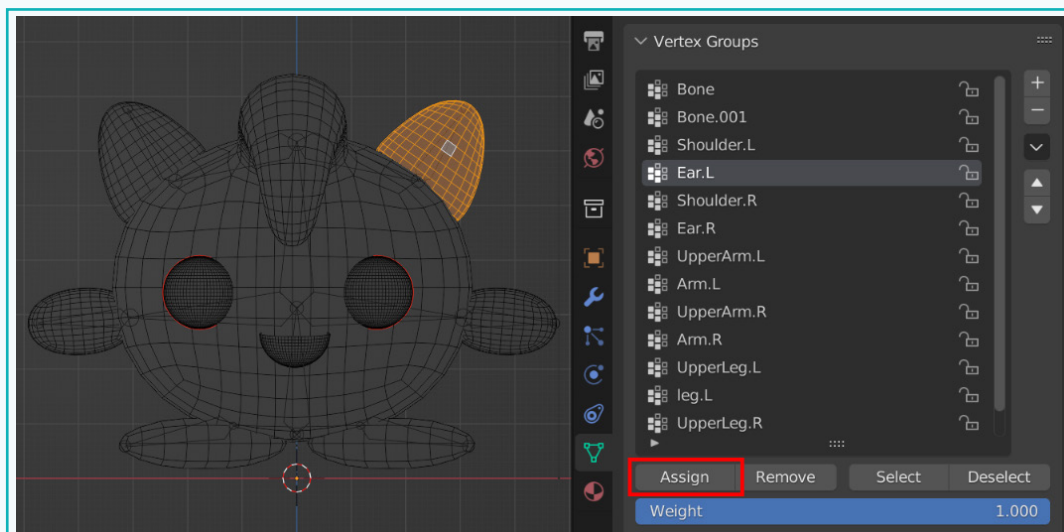
Activity

- 19 Press 'L' to select all the linked vertices/edges/faces. All the vertices on the ear get selected.



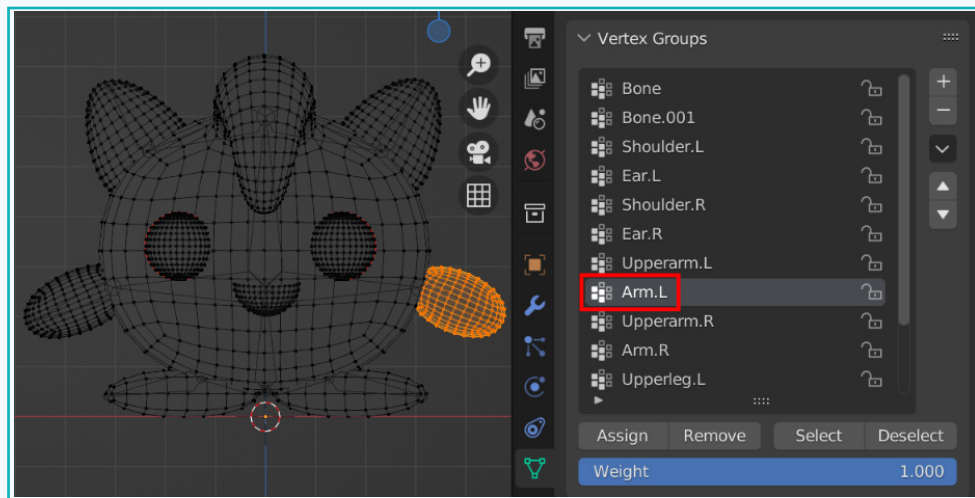
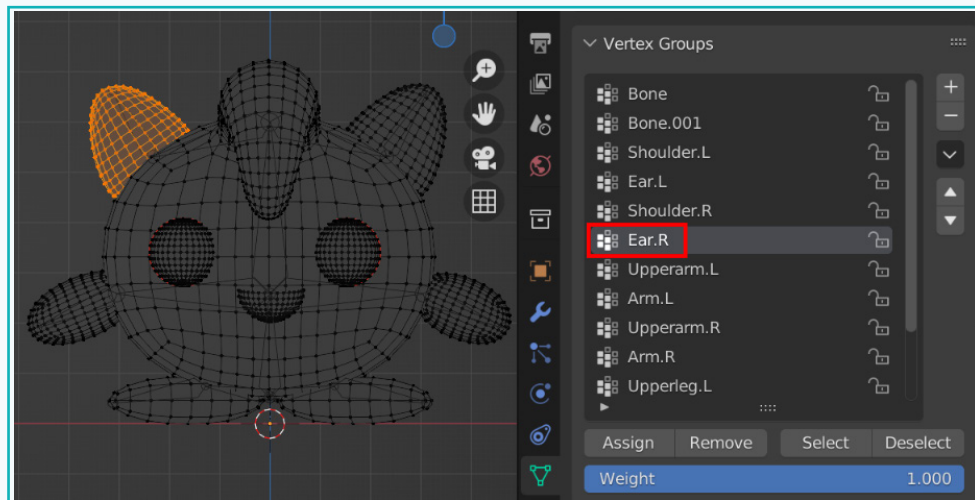
- 20 Now, click on the 'Assign' button. This assigns all the selected vertices/edges/faces to the selected vertex group.

While assigning, we can specify the exact weight value, under the 'Assign' button. A weight value of '1' indicates that these selected vertices will be 100% influenced by the 'Ear.L' joint/vertex group. We will assign a value of '1'.

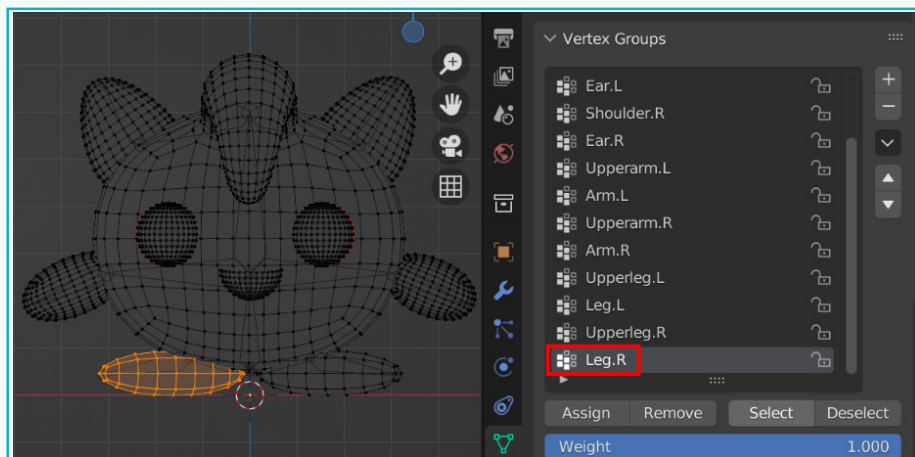
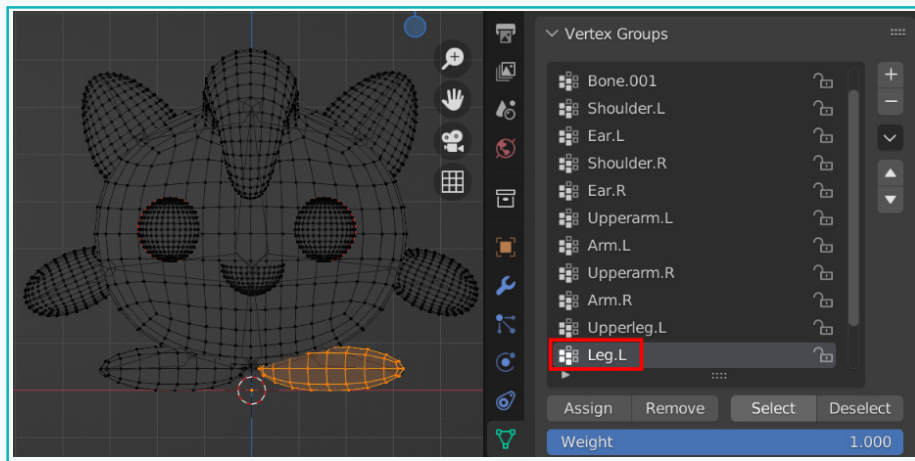
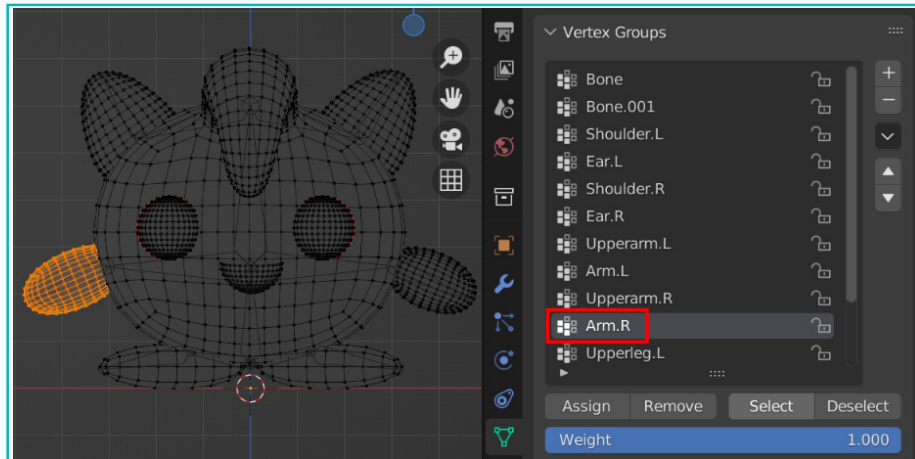


Activity

- 21** Follow the same steps for the other vertex groups. For simplicity, let us do it only for the parts that we are interested in animating – hands, feet, and ears.
- Select the vertex group and click on the ‘Select’ button.
 - Click on ‘Remove’.
 - Click anywhere in the 3D viewport to deselect the vertices.
 - Select one or more vertex/edges/faces on the related part of the body, then press ‘L’ to select all the linked elements.
 - Click on ‘Assign’.

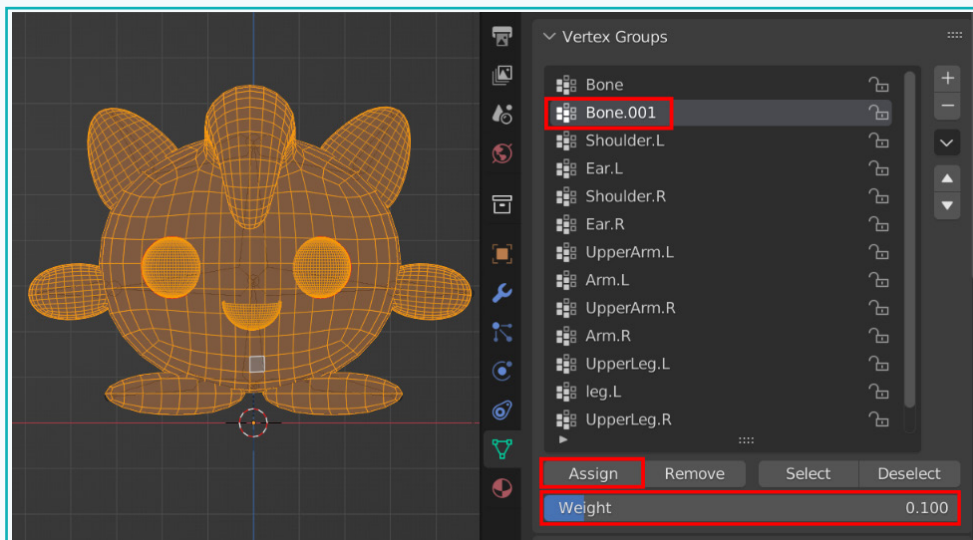
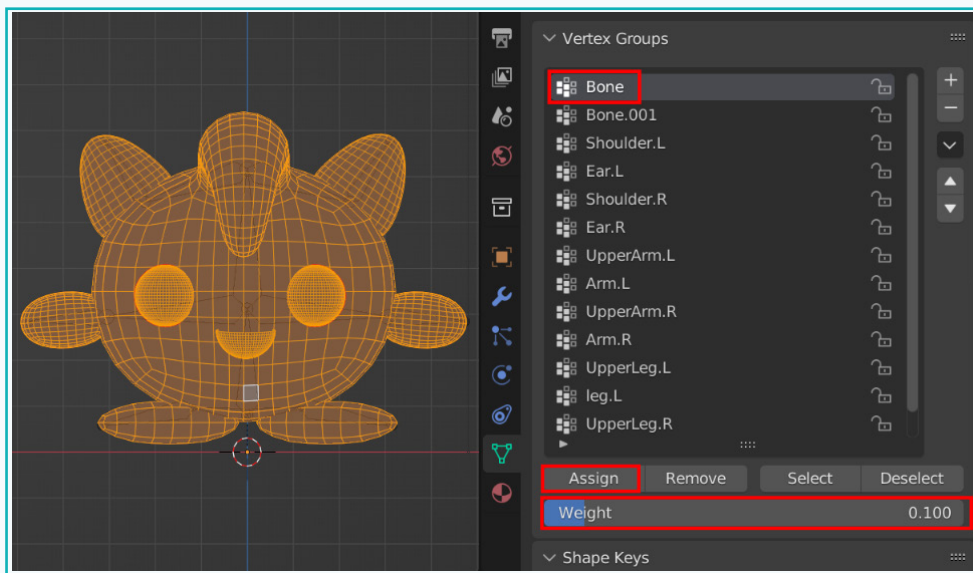


Activity



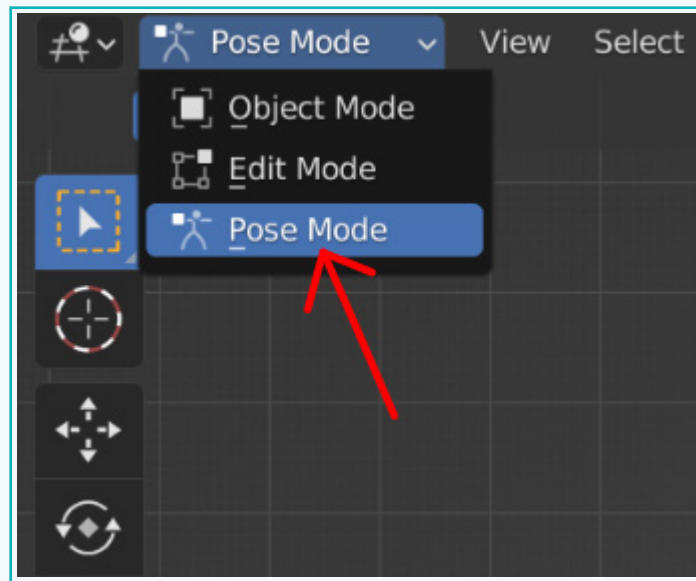
Activity

- 22 Also, for the spine bones, let us assign a weight of '0.1' to all the vertices of the character.

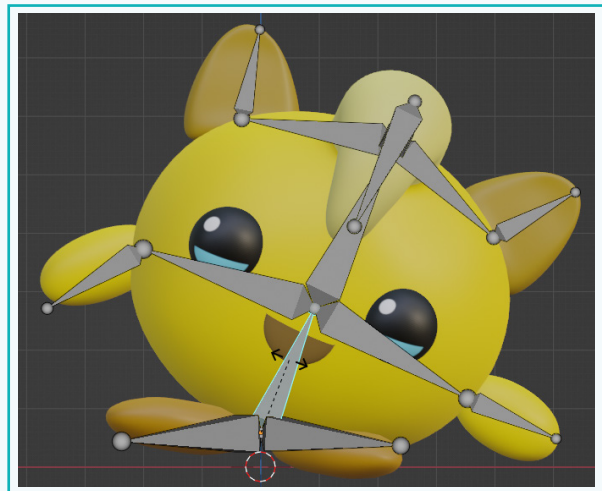
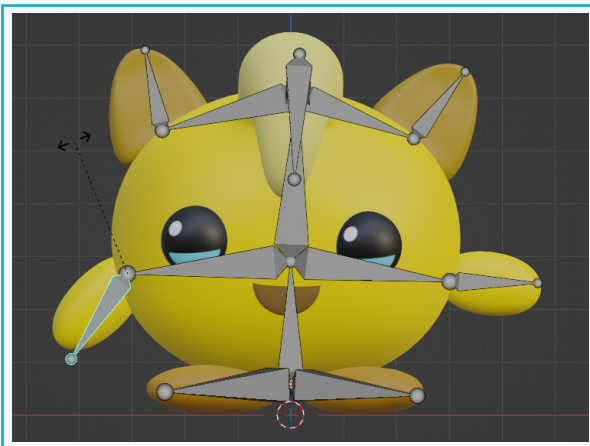


Activity

Once the vertices have been allocated for all the bones, go into the 'Object' mode, and select the armature. Then, go into the 'Pose' mode.

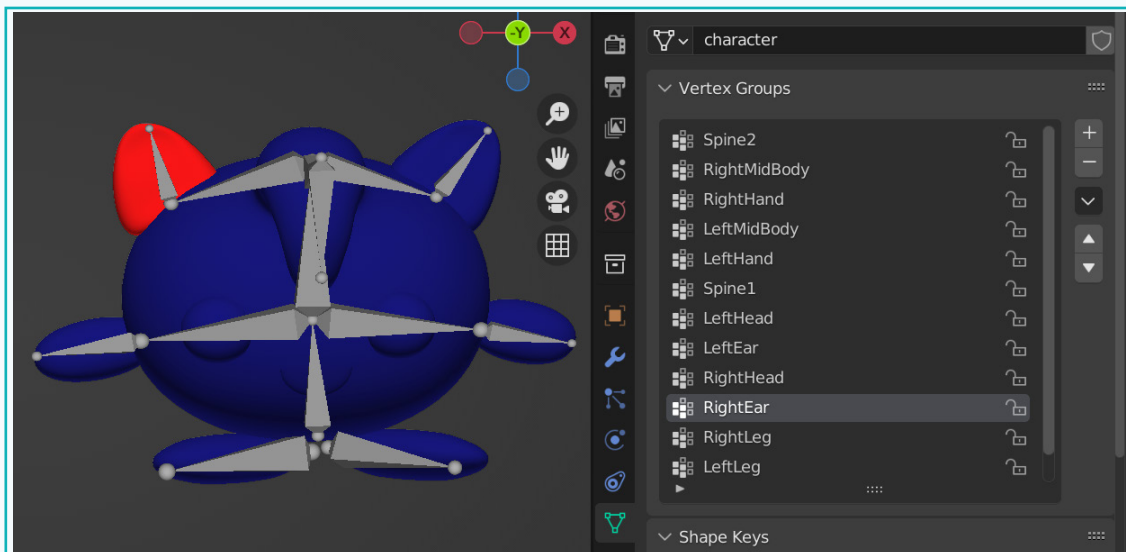
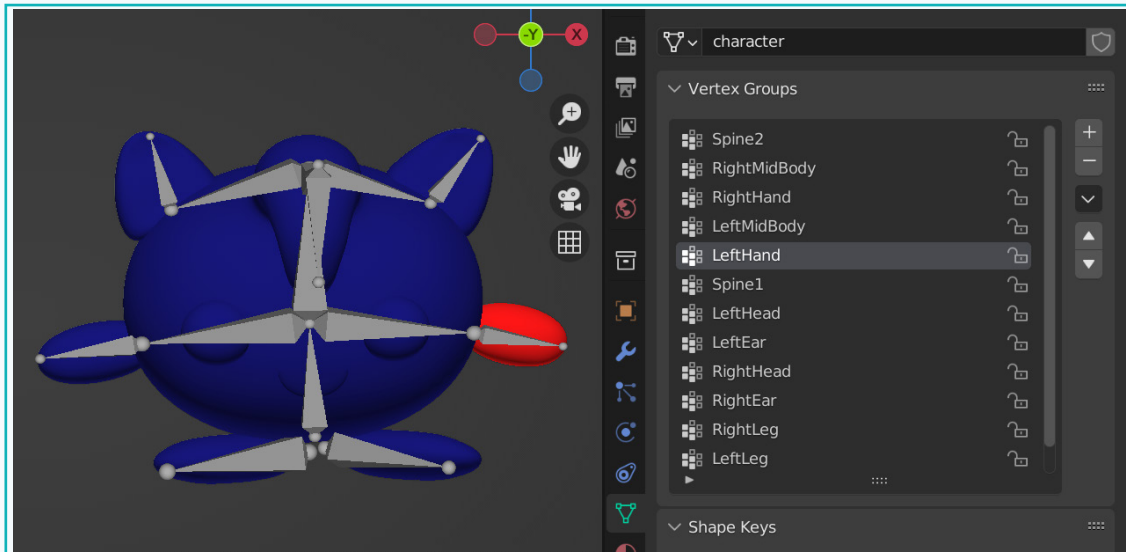


- 23 Move each bone and check if the correct part of the mesh is getting deformed.



Activity

- 24 You can also go into the 'Weight Paint' mode and check the vertex allocation.

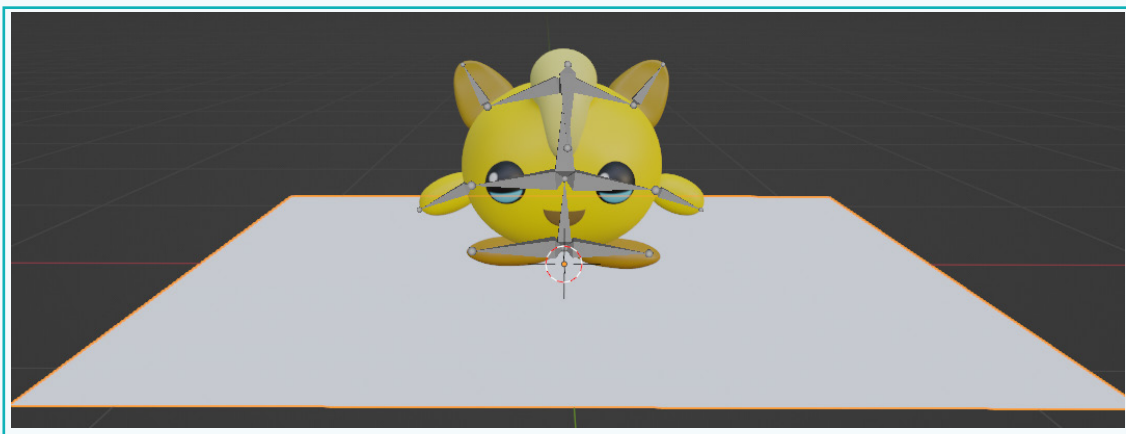


Now that the weight allocation for the bones is done, let us set up the environment for the animation.

Activity

Create Environment

- 1 Create a plane.

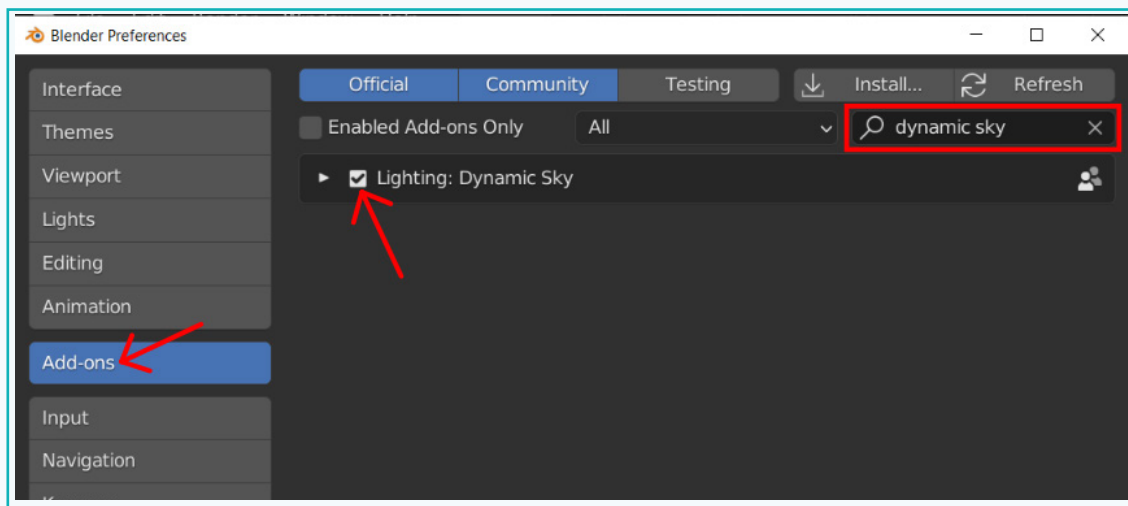


- 2 Change the colour of the plane and light up the scene using the 3-point lighting system or as desired.



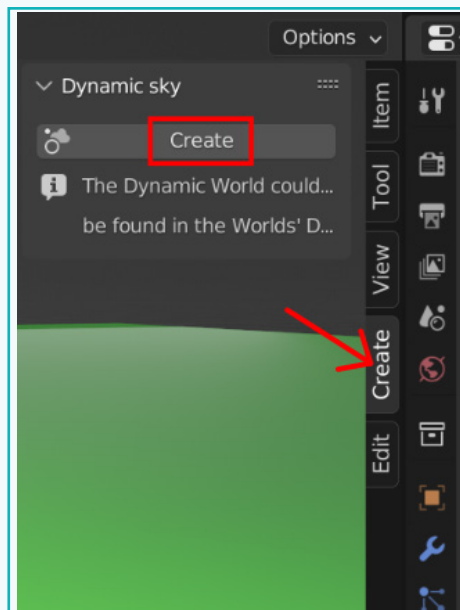
Activity

- 3 Let us create a sky background for our scene by making use of an add-on provided by Blender – **Dynamic Sky**. Go to 'Edit' → 'Preferences' → 'Add-ons'. Type 'dynamic sky' in the search box and enable the 'Add-ons' from the list below.



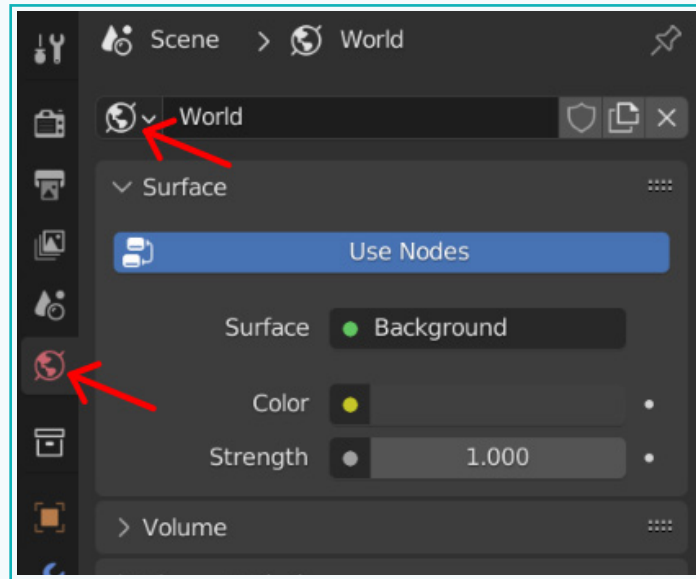
Once enabled, a new tab called 'Create' appears in the side panel area.

- 4 Click on the 'Create' tab. Then, click on the 'Create' button under the 'Dynamic Sky' dropdown.

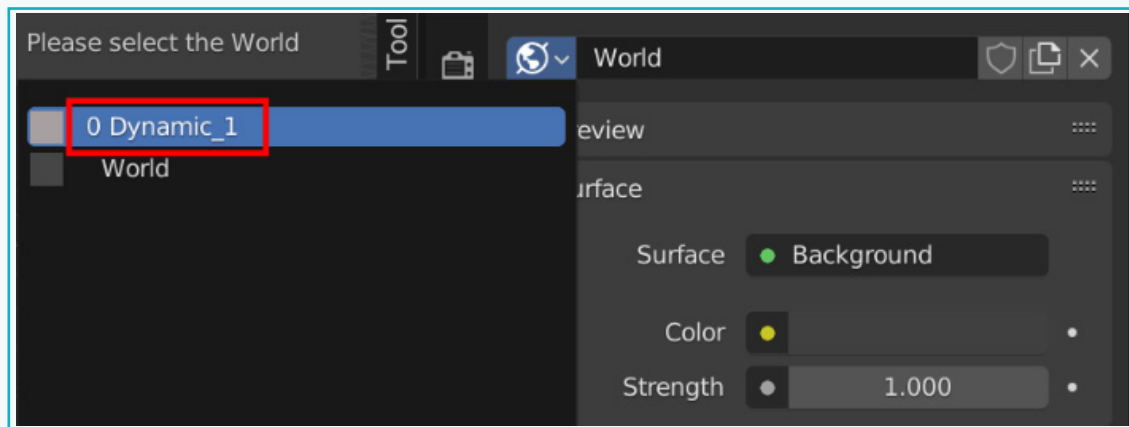


Activity

- 5 Next, go into the 'World' properties window and click on the dropdown next to the name of the world.

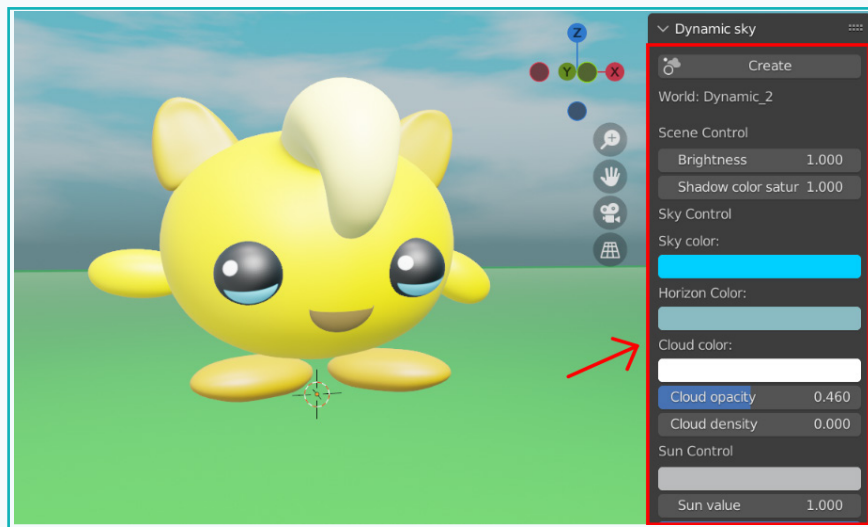


- 6 Select 'Dynamic_1' from the list.



Activity

- 7 The sky gets created in the background of our scene. The appearance of the sky can be modified through the properties on the side panel.

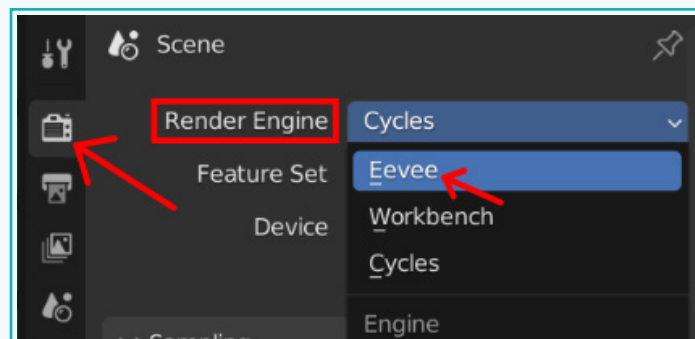


Also, if the display of the sky causes Blender to slow down or gets glitchy, consider shifting to the 'Eevee' render engine instead of 'Cycles'.

Render Engine

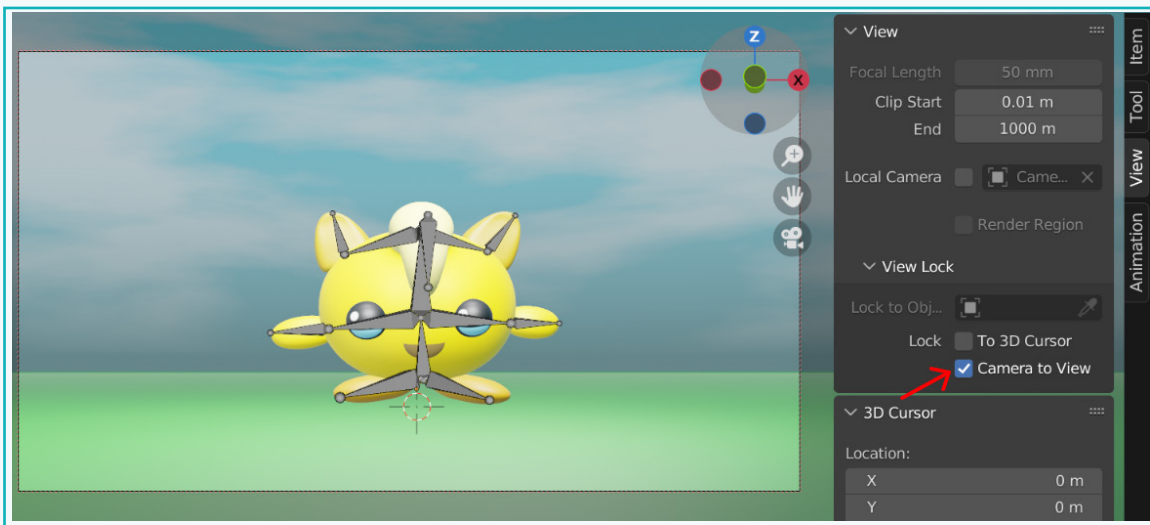
The **Render Engine** determines how our scene will look. 'Eevee' and 'Cycles' are both rendering engines used in Blender. While they both serve the purpose of rendering scenes, there are several key differences between Eevee and Cycles.

'Eevee' stands for 'Extra Easy Virtual Environment'. It is designed for real-time rendering and provides fast and interactive feedback, making it ideal for tasks requiring quick previews. On the other hand, 'Cycles' focuses on achieving high-quality, photorealistic results at the cost of longer rendering times.



Activity

- 8 Go to the 'View' tab in the side panel and enable the 'Camera to View' option. Position the character appropriately with respect to the camera.



Now that the joints and the environment are all set up, we can begin the process of adding keyframes to create an animation.

Keyframing

Keyframing refers to the process of creating and manipulating keyframes to animate objects or properties within a scene. A **keyframe** represents a specific point in time at which we set a value or a state for an object or a property. Blender automatically interpolates the values between keyframes to create smooth animations.

For example, we can set a keyframe for the position of an object at frame 1, then set another keyframe at frame 100 with a different position. Blender will automatically calculate the movement of the object between these keyframes, creating a smooth animation.

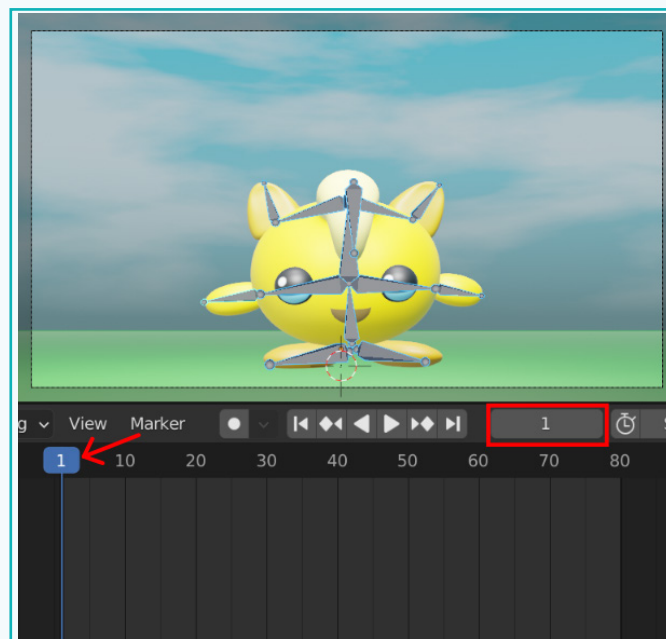
Keyframes can be specified for various properties in Blender such as object transformations (such as position, rotation, and scale), material settings, visibility, and more. By establishing and altering keyframes over time, we can create complex animations, character movements, etc.

Activity

- 1 Open the timeline editor and set the start and end keyframes of the animation.

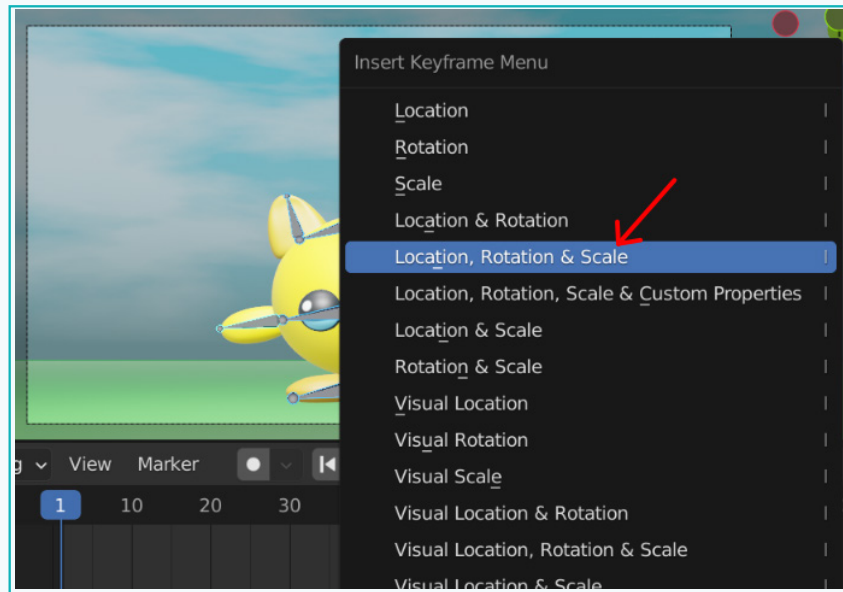


- 2 Set the marker at frame 1. Select the armature and go into the 'Pose' mode. Press 'A' to select all the bones.

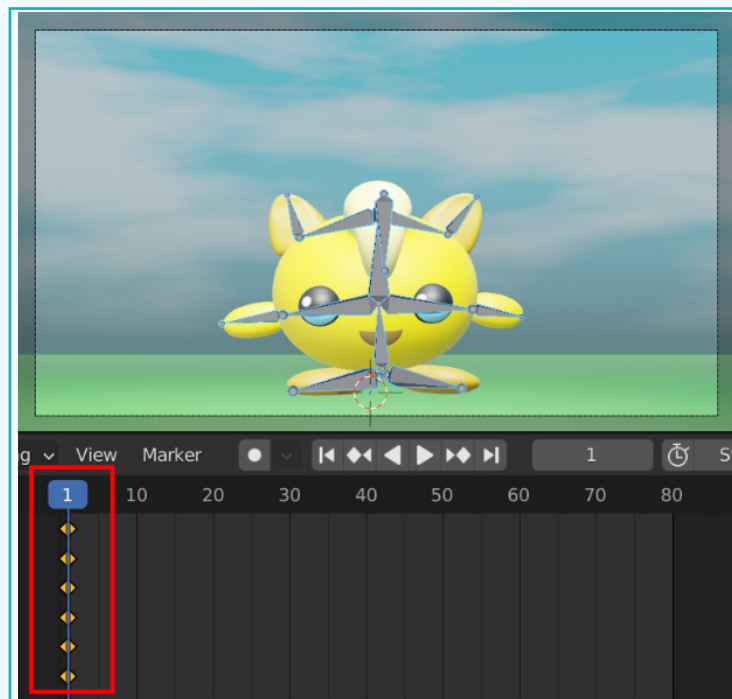


Activity

- 3 Press 'I' to display the 'Insert Keyframe' menu and select 'Location, Rotation & Scale' from the menu.

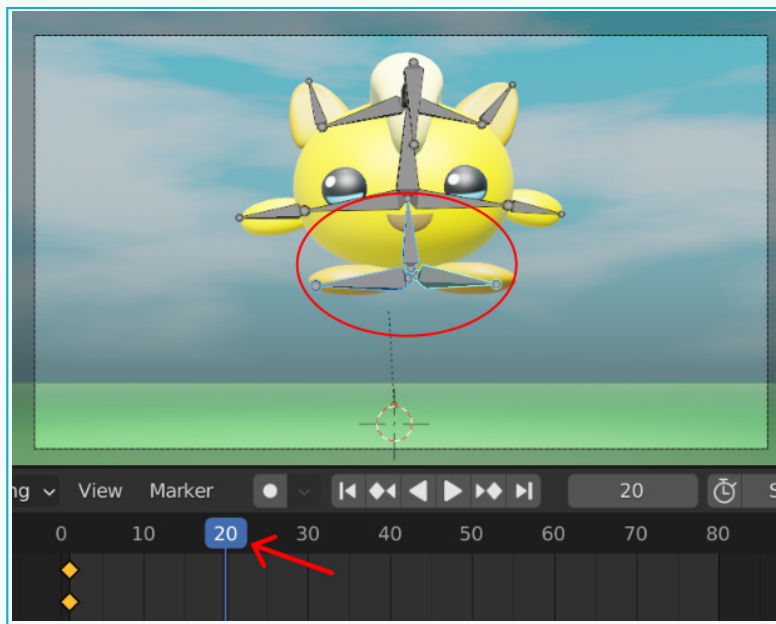


Keyframes get added at frame 1. This is the starting pose for the character.



Activity

- 4 Next, move the marker to frame 20. Select the three bones shown in the image below and press 'G' and move the character upwards.

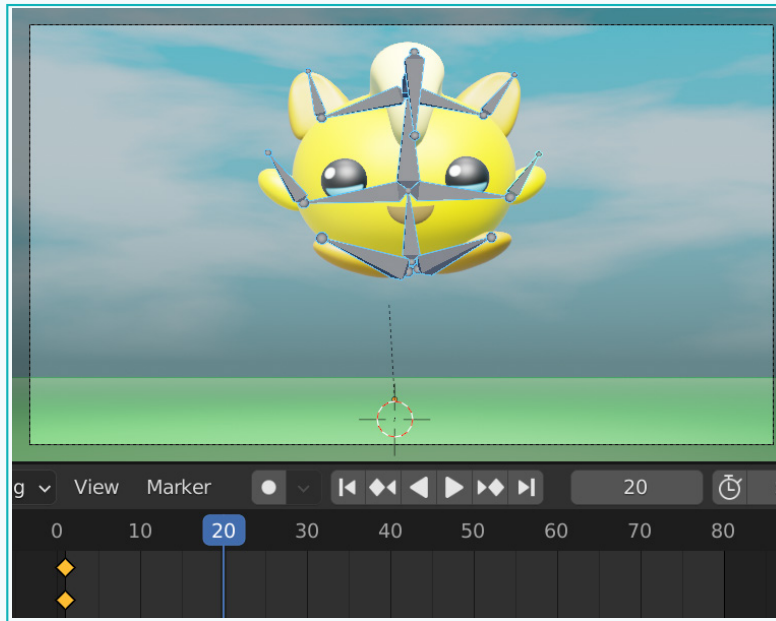


- 5 Also, move the hands and feet of the character upwards. Select the left-hand bone, then press 'R' to rotate and rotate the hand upwards. Repeat for the right hand, left foot, and right foot.

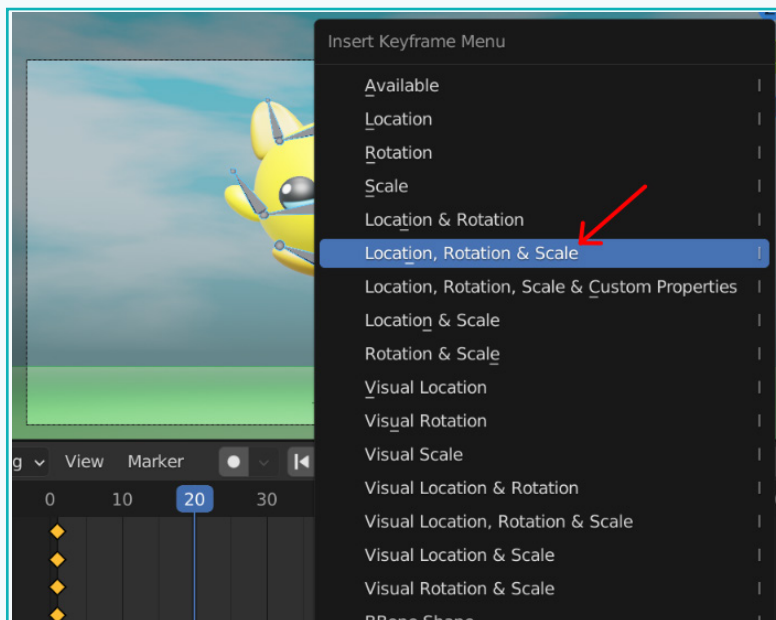


Activity

- 6 Next, press 'A' to select all the bones. This is because we must add keyframes for every bone that has been moved, rotated, scaled, or manipulated in any way.

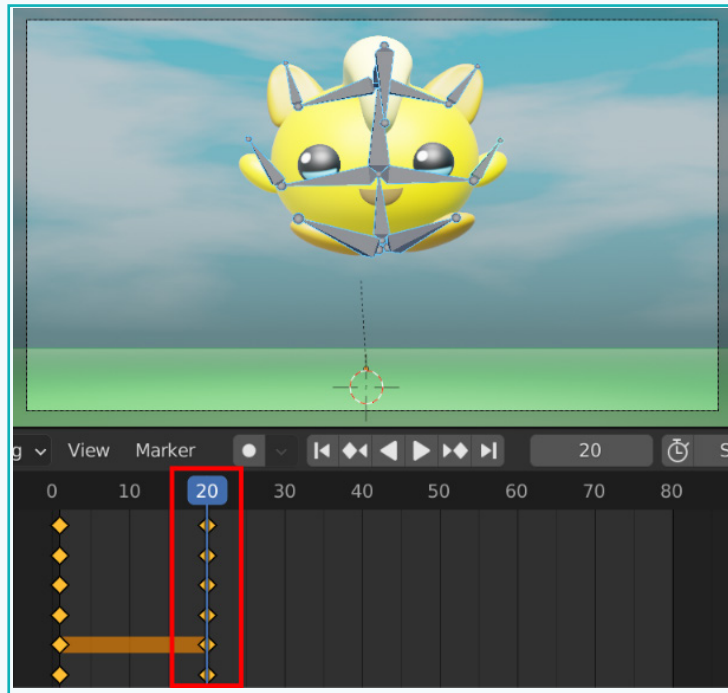


- 7 Press 'I' and add keyframes for 'Location, Rotation & Scale'.

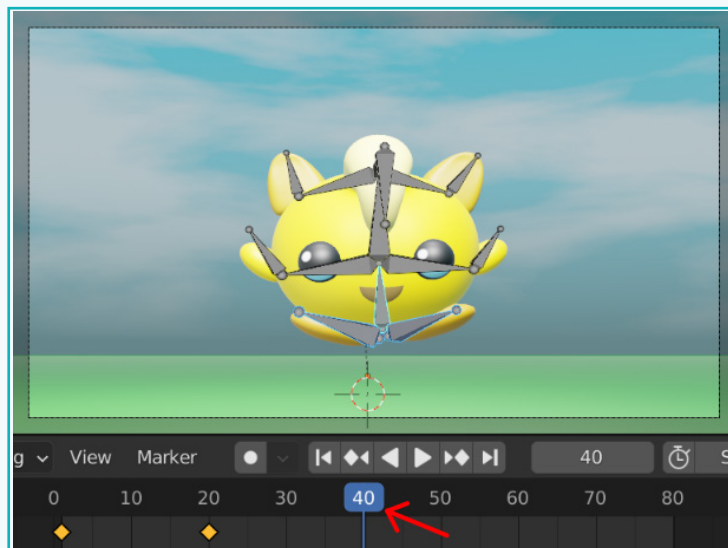


Activity

- 8 Keyframes get added at frame 20. Move the marker to 'keyframe 1' and click on 'Play' to test the animation.

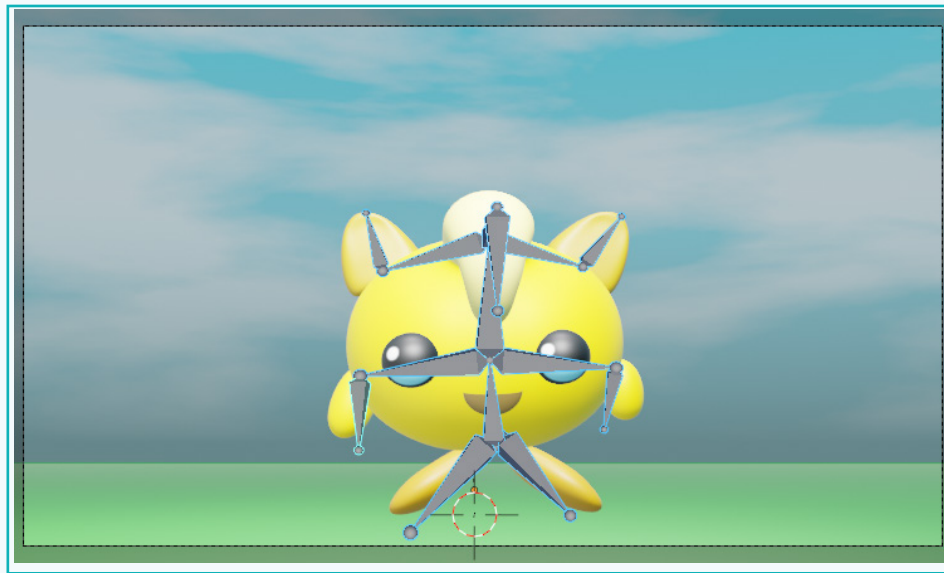


- 9 Next, we will add movement to allow the character to hop down again and move the marker at frame 40. With the same bones selected, move the model towards the ground.

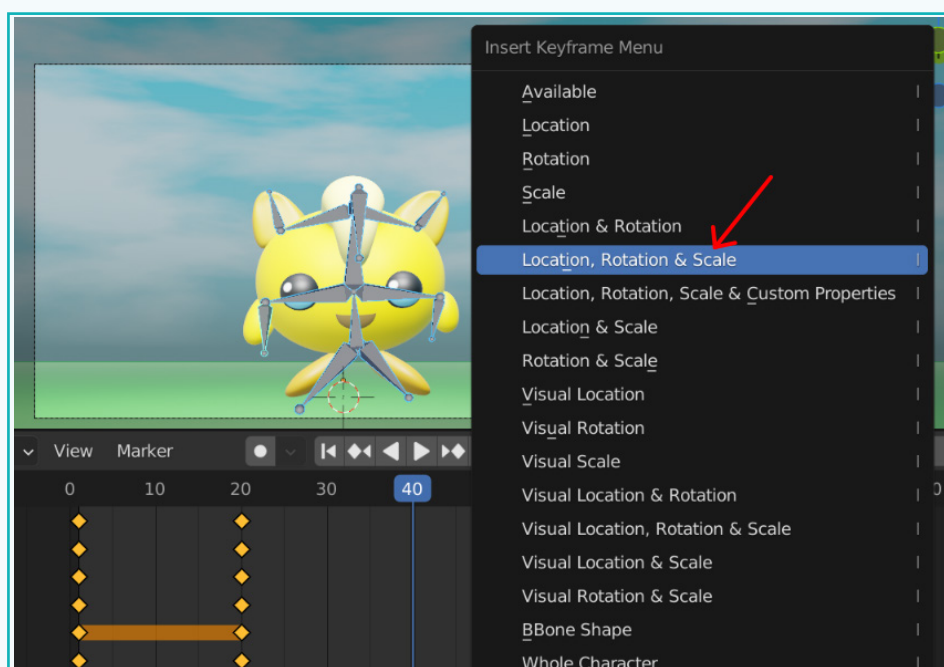


Activity

- 10 Also, move the hands and feet downwards. Once the pose is final, press 'A' to select all the bones.

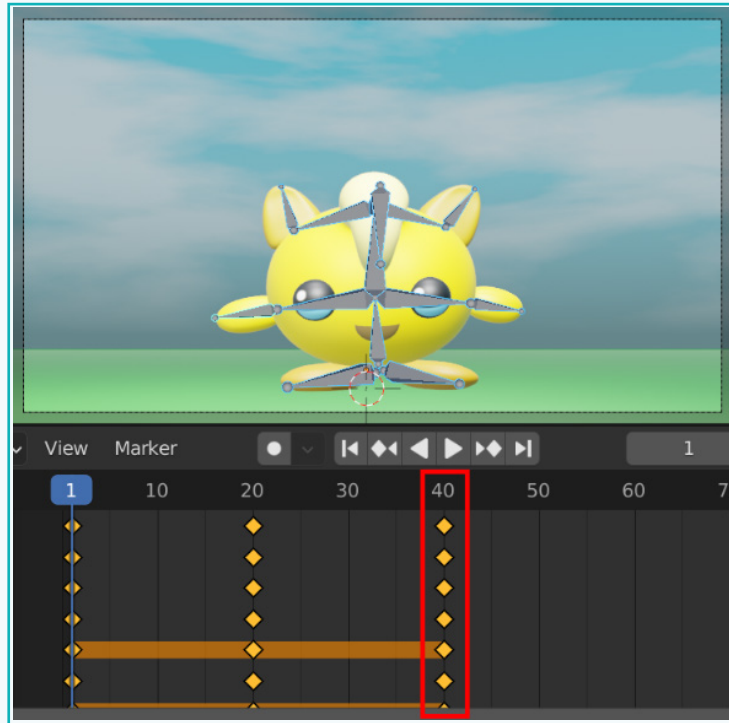


- 11 Press 'I' for the 'Insert Keyframe' menu and select 'Location, Rotation & Scale' from the menu.



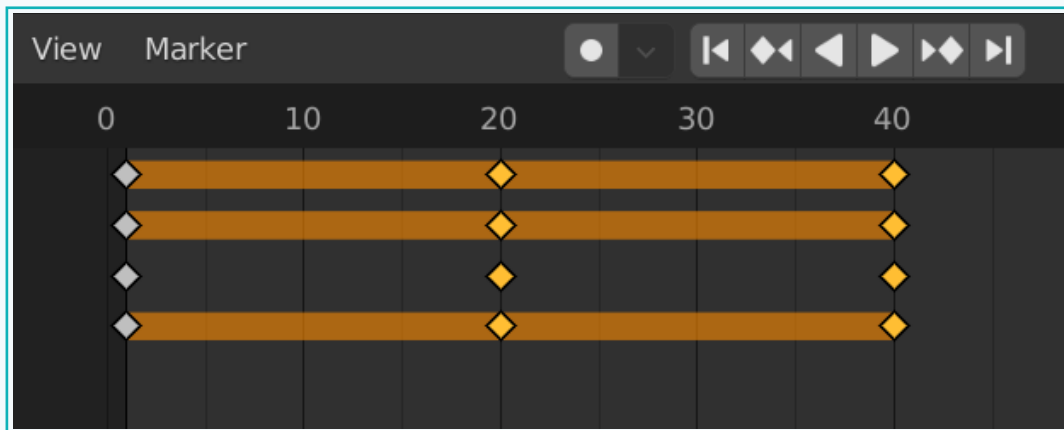
Activity

- 12 Keyframes get added at frame 40. Click on 'Play' and check the animation.



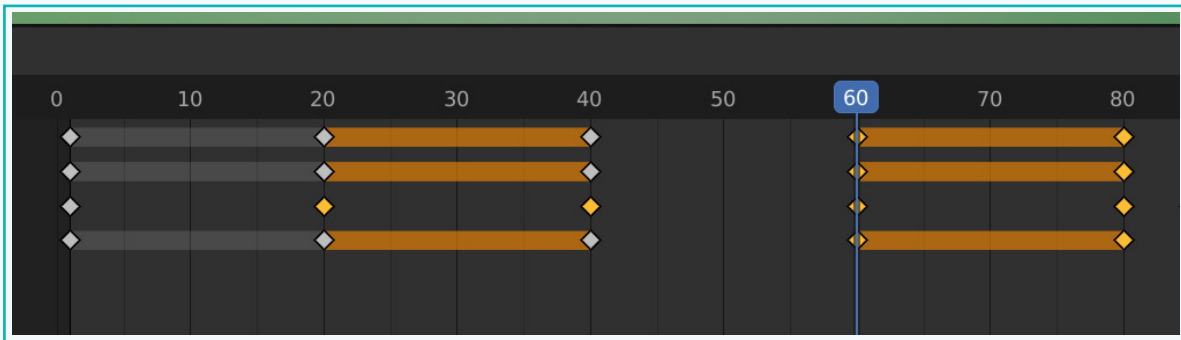
Now, to reduce efforts, we will duplicate the keyframes present at frames 20 and 40 and move them to frames 60 and 80, respectively.

- 13 Using 'Box Select', select all the keyframes at frames 20 and 40.



Activity

- 14 Press 'Shift + D' to duplicate the selected keyframes. Move them to the frames 60 and 80.



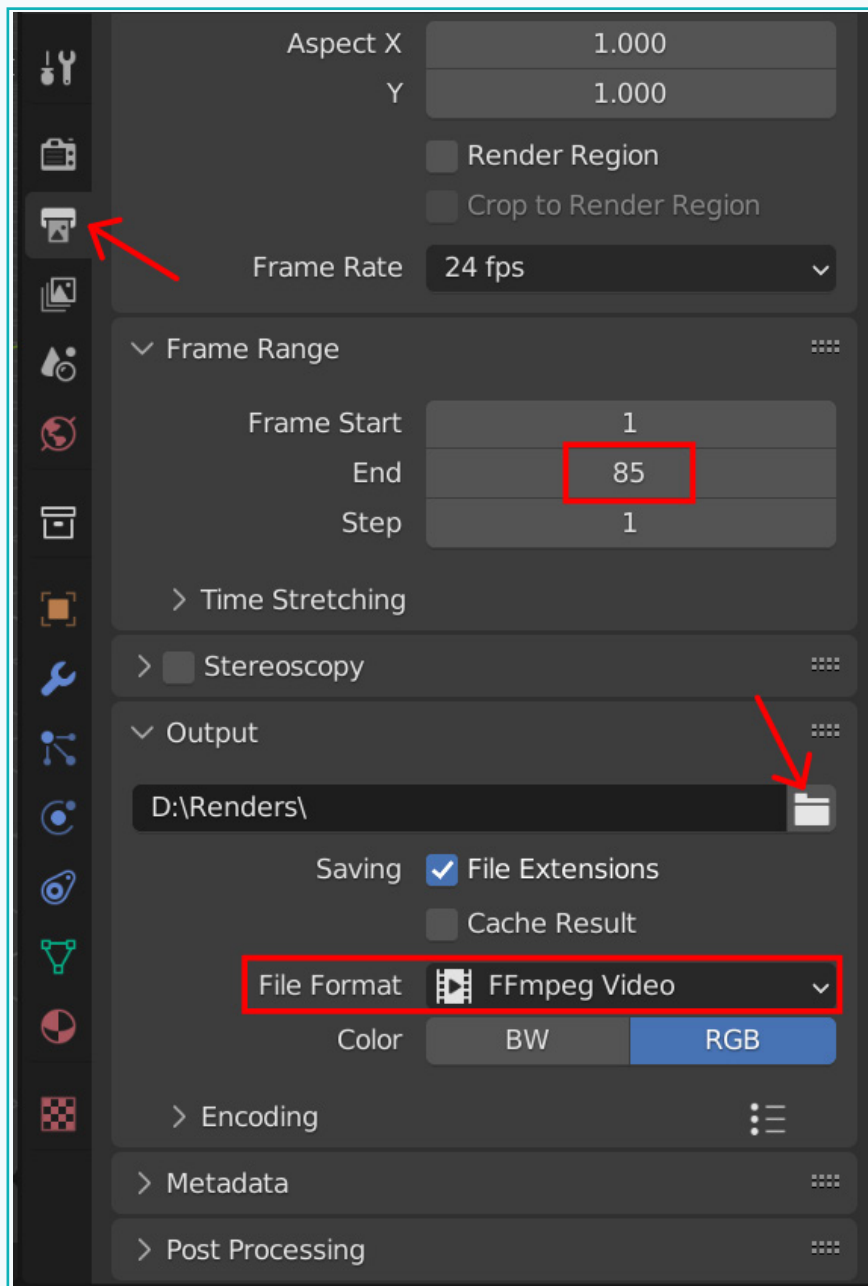
Try some more modifications, like changing the scale of the character's body while moving up/down. Click on 'Play' and check the output.

Rendering the Animation

Once we are done with the keyframes placement and animation, we can start with the process of rendering the animation. First, some configuration is required.

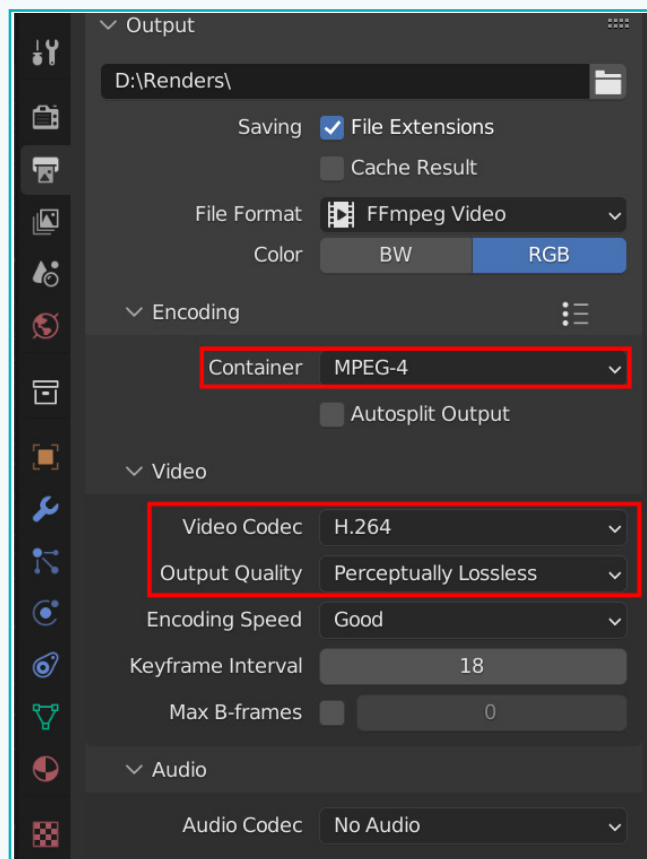
Activity

- 1 Click on the 'Output Properties' panel in the 'Properties' window. Under 'Frame Range', set the 'End' value to '85'. Under 'Output', set the output folder path. This is the location where the animation video will be saved. Set the file format to 'FFmpeg video'.

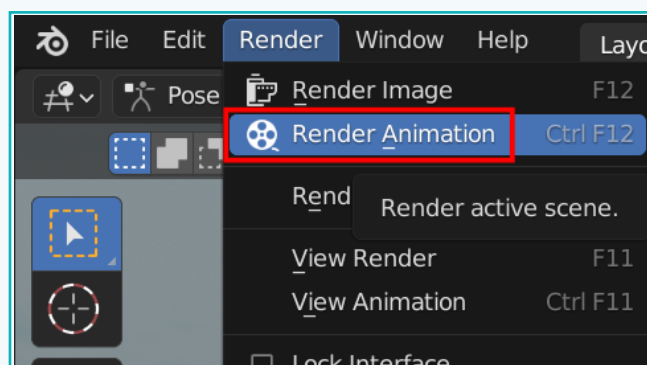


Activity

- 2 • Under 'Encoding', set 'Container' to 'MPEG-4'.
 - Under 'Video', set 'Video Codec' to 'H.264' and 'Output Quality' to 'Perpetually Lossless'



- 3 We are now set to render the animation. Click on the 'Render' menu → 'Render Animation'.



The 'Render' window will open, and the animation will start getting rendered into the given destination folder.

In this lesson, we learnt about the various aspects involved in animating a 3D character. We explored the different stages of the character animation process like rigging and keyframing.

Exercise

Fill in the blanks.

- 1 The _____ allows the character model to be in a symmetrical position with its arms outstretched and legs together.
- 2 The _____ tool only works if the bones are named using the proper naming convention.
- 3 _____ involves associating the character's mesh with the joints.
- 4 The process of assigning weight values to each joint's influence on the vertices is known as Weight _____ or Weight _____.
- 5 In the Weight Paint mode, _____ represents a weight of 0 or no influence at all.
- 6 Eevee stands for _____.
- 7 A _____ represents a specific point in time where we set a value or a state for an object or a property.

Exercise

 **State if the statements are True or False**

8 Eevee focuses on achieving high-quality, photorealistic results at the cost of longer rendering times.

 _____

9 Rigging is the process of creating a digital skeleton or structure for the character.

 _____

10 The first step in the rigging process is Joint Hierarchy.

 _____

10. Character Animation

Hope you enjoyed the lesson. We have explored a specialised area of the animation process. Using character animation, you can now breathe life into 3D characters, adding nuance and personality through their movements, gestures and facial expressions.



Tech Flash

- **Storyboarding** is the process of planning out the sequence of actions and movements that the character will perform.
- **Rigging** is the process of creating a digital skeleton or structure for the character.
- **Keyframing** is a technique used to define the important poses and movements in animation.
- **Joint Placement** is the process of placing joints in the character's model.
- **Binding** is the process of binding the character's mesh with the joints.
- **Weight Painting/Allocation** is the process of assigning weight values to each joint's influence on the vertices.